Supplemental Material for
“Reading a magnet’s Hamiltonian term by term: two-dimensional terahertz spectroscopy as vertex spectroscopy of $TmFeO_3$”

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Contents

S1 The SU(3) qutrit and why its classical dynamics is quantum-exact

Tm^{3+} is $4f^{12}$, ${}^3H_6$ ($J = 6$). In the $C_s(m)$ crystal field of the *Pbnm* 4*c* site the 13-fold multiplet splits into 13 singlets; the model keeps the three lowest, $|1\rangle, |2\rangle, |3\rangle$, with splittings $E_{12} = 2.068\,\mathrm{meV}$ ($0.50\,\mathrm{THz}$) and $E_{13} = 4.963\,\mathrm{meV}$ ($1.20\,\mathrm{THz}$), consistent with the measured lines $E_{12} + E_{23} \approx E_{13}$. The site state is the Gell-Mann vector $\lambda^a = \mathrm{Tr}(\rho\Lambda^a)$, $a = 1\dots 8$, and every single-ion term (CEF energies, external drives, exchange mean fields) is *linear* in the Λ^a. For a Hamiltonian linear in the generators the von Neumann equation $\dot\rho = -i[H,\rho]$ closes on $\vec\lambda$:

$$\dot\lambda^a \;=\; \Omega_{ab}(t)\,\lambda^b - \gamma_a\big(\lambda^a - \lambda^a_{\mathrm{eq}}\big), \qquad \Omega_{ab} = \tfrac12\,\mathrm{Tr}\Big(\Lambda^a\,(-i)[H(t),\Lambda^b]\Big)\,. \tag{1}$$

The classical SU(3) propagation is therefore the *exact* quantum dynamics of the driven, damped three-level ion, and a Boltzmann-weighted ensemble of trajectories launched from the level seeds $|1\rangle, |2\rangle, |3\rangle$ is the exact thermal density matrix for single-ion drives. (This was verified explicitly against an independent density-matrix integrator; the two censuses agree to all digits quoted.) Damping is Bloch-type on the Gell-Mann pairs of each transition, $\gamma(\lambda^{1,2}), \gamma(\lambda^{4,5}), \gamma(\lambda^{6,7})$, with population relaxation $\gamma(\lambda^{3,8})$ toward the equilibrium populations captured at initialization.

The magnetically bright coordinate of the E_{12} block follows from the projected moment $PJ_zP = 5.264\,\lambda^2$: only λ^2 radiates magnetically, and the onsite splitting drives the planar precession $\dot\lambda^1 = -\Delta_{12}\lambda^2$, $\dot\lambda^2 = +\Delta_{12}\lambda^1$, so the E_{12} sector is a single driven oscillator

$$\lambda^2(t) = \int_0^t \sin[\omega_{12}(t-t')]\,h^1(t')\,dt', \tag{2}$$

where h^1 is the effective force a conversion vertex exerts on λ^1. All delay-axis structure of a cross peak lives in the τ-dependence of h^1.

S2 Crystallography and the projected Fe–Tm coupling

$\mathrm{TmFeO_3}$ is *Pbnm* ($Z = 4$): Fe^{3+} at 4*b* (site symmetry $\bar 1$; every Fe sits on an inversion center), Tm^{3+} at 4*c* (site symmetry the single mirror m_z). Inversion maps each Fe sublattice to itself and exchanges Tm sublattices in pairs; the local mirror is the physical 4*c* site mirror and generates the relative λ^5, λ^7 signs between Tm sublattices. The Fe order is G-type ($T_N \approx 632\,\mathrm{K}$) with DM canting; below the spin reorientation ($T_2 \approx 85\,\mathrm{K}$) the ground state is $\Gamma_2 = (F_x, C_y, G_z)$.

The Fe–Tm sector is built from the physical operator route

$$\boldsymbol S,\ \boldsymbol J \;\longrightarrow\; H_{\mathrm{Fe-Tm}} = \boldsymbol S^T A\,\boldsymbol J \;\longrightarrow\; P\boldsymbol J P \;\longrightarrow\; \text{couplings to } \lambda^a, \tag{3}$$

never by assuming that a space-group rotation acts as a closed onsite SU(3) rotation in one fixed qutrit basis. Working in a *transported gauge* (qutrit bases on the four Tm sublattices related by the group operations themselves) makes every bond formula explicit and turns the irrep (Γ_1) projection into simple four-site sign sums. Time-reversal splits the Gell-Mann set into $\lambda^- = \{\lambda^2, \lambda^5, \lambda^7\}$ (odd) and $\lambda^+ = \{\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8\}$ (even); every coupling term must be $\mathcal{T}$-even overall, which fixes how many Fe-spin or B-field legs may attach to each sector. The candidate couplings through cubic order in the fields are

family	operator form	sector	legs
K^-	$S_\alpha \lambda^a$	λ^-	1 Fe
ESJ	$E_\eta S_\alpha \lambda^a$	λ^-	1 E-photon + 1 Fe
κ^B	$B_\eta S_\alpha \lambda^a$	λ^+	1 B-photon + 1 Fe
W	$S_\alpha S_\beta \lambda^a$	λ^+	2 Fe

The full orbit formulas (all eight coefficients per bond, all four Tm sublattices, both time-reversal sectors) and the implementation prescriptions for global-axis versus local-frame storage are given in the long-form working note that accompanies the data (`tmfeo3_foundation.tex`, Secs. 3–14).

S3 The sharp requirement and the vertex scorecard

Two collinear pulses give $M_{\mathrm{NL}} = M_{AB} - M_A - M_B$; order by order only the mixed terms survive, so M_{NL} starts at second order and the leading robust magnetic-dipole contribution is third order. Combining Eq. (2) with the conservation rules at each vertex (energy; $\mathcal{T}$-parity; Γ_1) yields six requirements for a coupling to produce the geometry-A cross peak at $(q_{\mathrm{AFM}}, E_{12})$:

R1 (detection) reaches λ^1/λ^2 (the E_{12} block);

R2 (drive) is sourced by the pumped δS_y magnon;

R3 (parity) is $\mathcal{T}$-even (fixes the $\lambda^\pm$ sector);

R4 (irrep) is Γ_1-allowed in the Γ_2 state;

R5 (energy) carries the $0.90 \to 0.50$ THz mismatch (needs ≥ 2 non-λ legs);

R6 (M_{NL}) involves both pulses.

Table S1: Vertex scorecard. Only W and κ^B pass all six requirements.

vertex	legs	sector	R1	R2–R6	verdict
K^-	$1\mathrm{Fe} + 1\mathrm{Tm}^-$	λ^-	×	×	hybridizes only; Γ_1-blocked; reorients Γ_2
ESJ	$E + \mathrm{Fe} + \mathrm{Tm}^-$	λ^-	×	✓	real, but feeds E_{13}/E_{23}, not E_{12}
W	$2\mathrm{Fe} + \mathrm{Tm}^+$	λ^+	✓	✓	**winner**: rank-1, factorized peak
κ^B	$B + \mathrm{Fe} + \mathrm{Tm}^+$	λ^+	✓	✓	works; B_z-gated $\Rightarrow$ geometry-B vertex

K^- **fails threefold.** (i) Linear in S, it only rotates the harmonic normal-mode basis: coupled harmonic oscillators give diagonal peaks only; a static bilinear conserves energy and cannot convert 0.90 into 0.50 THz. (ii) The Γ_1 orbit sums vanish for every channel except K^-_{2y}: a static one-at-a-time scan at 0.2 meV leaves the energy, the Fe order, and the uniform $(\lambda^2, \lambda^5, \lambda^7)$ *bit-identical* to baseline for all components except $2y$. (iii) The one allowed channel K^-_{2y} is a transverse molecular field on the order parameter and reorients $\Gamma_2 \to G_y$ between 0.085 and 0.090 meV—it is not a perturbative vertex. Full-dynamics 2DCS on the inert channels (K^-_{2x} up to 1.5 meV, K^-_{2z} up to 2.0 meV) leaves the induced λ^2 at machine precision ($\leq 7 \times 10^{-15}$).

The two winners. The two-magnon vertex rectifies the pulse-A/pulse-B cross term,

$$h^1_{\rm NL}(t,\tau) = W_1^{yy} A^2 \left[\cos(q_{\rm AFM}\tau) - \cos(q_{\rm AFM}(2t+\tau))\right] \theta(t) \;\Rightarrow\; M^W_{\rm NL} \propto \cos(q_{\rm AFM}\tau)\left[1 - \cos\omega_{12} t\right], \quad (4)$$

a clean factorized peak with no diagonal partner. The field-gated vertex kicks at the probe, $h^1 \propto B_y(t) S_y(t)$ with $S_y(0) = A \sin q_{\rm AFM}\tau$, giving $M^\kappa_{\rm NL} \propto \sin(q_{\rm AFM}\tau)[1 - \cos\omega_{12} t]$: also on target, but the same gate dumps each pulse's local response into Tm at both pulse times, generating an (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background. Its dominant symmetry slice is z-gated ($B_z(t)$), so in geometry A ($B_z = 0$) it is *identically silent*: the gate is simultaneously the reason it is harmless in geometry A and active in geometry B.

Exhaustion of the quadratic families. All remaining symmetry-allowed quadratic couplings were switched on one at a time in the full nonlinear dynamics and are exactly inert (differences at the 10^{-12} level or below): $W_4^{yy,xy,yz,xx,zz,xz}$, W_1^{xy}, $W_6^{yy,xz}$, κ^E_{2y}, κ^E_{5x}. Among the W_1 components: W_1^{yz} inert, W_1^{zz} negligible, W_1^{xx} alive but floods the spectrum with an $\omega_t = 0.16$ THz comb, W_1^{xz} alive and on-target for hybridization (used below), and no static component can replace the gated κ^B for the geometry-B transfer peak: the entire $W_1 + W_4$ family fails to produce $(E_{12}, q_{\rm AFM})$ in geometry B without simultaneously contaminating geometry A. The B_z-gate is therefore *proven necessary by exhaustion.*

S4 Numerical protocol

Dynamics. Classical LLG for Fe (unit spins, $S = 5/2$ scale absorbed), SU(3) Bloch dynamics for the four Tm sublattices; RK4 with $\Delta t = 0.02$ code units; $1 \times 1 \times 1$ cell of the *Pbnm* magnetic unit (8 Fe + 4 Tm). Time is measured in $\hbar$/meV; a frequency f in THz corresponds to $E = f \times 4.136$ meV.

2DCS protocol. Delay scan $\tau \in [-120, 0]$ code units, $\Delta\tau = 0.1$ (0.2 for scans); detection window to $t_{\max} = 160$ for 0.01 THz resolution on ω_t. The nonlinear signal is the honest three-run subtraction $M_{\rm NL} = M_{AB} - M_A - M_B$ per delay; no pulse-window chunking; no reuse of the reference run across delays (both shortcuts were found to contaminate spectra and are documented pitfalls). The pump enters as a tabulated waveform (the measured field, times converted $t_{\rm code} = t_{\rm ps}/0.6582$, auto-centered and normalized).

Spectra and blind peak census. 2D spectra are $|{\rm FFT}_{\tau,t}|$ of $M_{\rm NL}$ with: detection window $t \geq 3$ (excludes pulse overlap), Gaussian t-apodization to 0.03, cosine tapers on both τ edges, global mean subtraction. The delay axis is plotted physically ($\omega_T = -\omega_\tau^{\rm FFT}$; $+ =$non-rephasing). Peaks are accepted only by a blind census: 2D local maxima (maximum filter, 0.10 THz neighborhood) with prominence ≥ 1.5 over the local median background (0.34 THz annulus) and amplitude $\geq$ 4–5% of the map maximum, merged within 0.09 THz. Rectified ($\omega_t \approx 0$) content is analyzed separately via

$S_{\rm DC}(\tau) = \langle M_{\rm NL}\rangle_t$ with cubic drift removal. In every 2D map the τ-averaged signal is subtracted at each detection time before windowing, which functionally removes the $\omega_\tau = 0$ (pump–probe) row; the slow-in-τ wings that remain within $|\omega_\tau| < 0.12\,$THz are masked. The model reproduces the observed dominance of the $(0, E_{23})$ band in the same-polarized channel before this removal; $\omega_\tau = 0$ features carry pump–probe, not delay-coherence, information and are excluded from all censuses. Radiation bookkeeping used throughout: for $k \parallel b$, M1 dipoles radiate $E \propto \hat{k} \times m$, so the $E \parallel c$-detected channel carries $d_z(\lambda^2) + m_x$ and the $E \parallel a$-detected channel carries $d_x(\lambda^{5,7}) + m_z$.

Thermal ensembles. Boltzmann mixtures of trajectories from the level seeds $|1\rangle, |2\rangle, |3\rangle$ (Sec. S1). Two documented pitfalls: (i) any pre-dynamics energy minimization collapses excited-level seeds and must be disabled; (ii) population damping $\gamma(\lambda^{3,8})$ relaxes toward the equilibrium captured at initialization, which must therefore be captured per-seed.

S5 Experimental inputs

Table S2: Measured mode parameters and the derived simulation inputs. CEF damping is Bloch on the Gell-Mann pair of each transition, $\gamma_a[\text{meV}] = \gamma[\text{THz}] \times h$; magnon damping is Gilbert, $\alpha = \gamma/2\nu$ fixed by the $q_{\rm AFM}$ line. Emission weight is $\sqrt{f}$ (E1) for the CEF lines and M1 for the magnons. The last column is the measured pulse spectral amplitude at each mode.

mode	ν [THz]	f	$\sqrt{f}$	γ [THz]	damping (sim)	pulse amp.
$q_{\rm FM}$	0.38	0.08	0.28	0.005	$\alpha = 1.7 \times 10^{-3}$	0.92
$q_{\rm AFM}$	0.90	7×10^{-4}	0.026	0.003	$\alpha = 1.7 \times 10^{-3}$	0.68
E_{12}	0.50	6.5	2.55	0.038	$\gamma(\lambda^{1,2}) = 0.157\,$meV	1.00
E_{23}	0.70	6.6	2.57	0.10	$\gamma(\lambda^{6,7}) = 0.414\,$meV	0.85
E_{13}	1.20	8.7	2.95	0.05	$\gamma(\lambda^{4,5}) = 0.207\,$meV	0.36

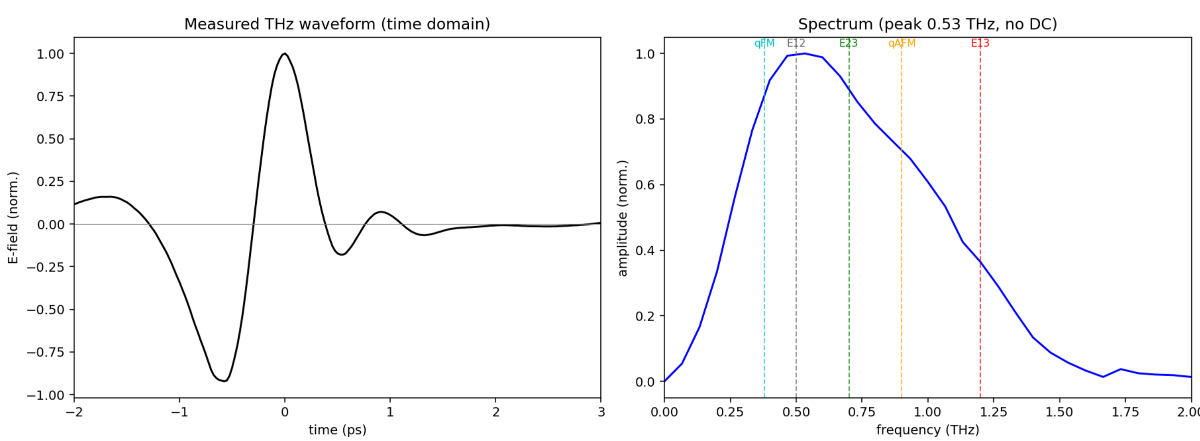

Figure S1: The measured THz pump. Left: single bipolar cycle ($\sim 1.2\,$ps, no DC). Right: amplitude spectrum peaked at $0.53\,\text{THz} \approx E_{12}$, covering all five modes. The absence of DC content removes the quasistatic-tilt legs that a Gaussian stand-in pulse would add; the measured waveform is the cleanest configuration.

Two consequences organize the readout physics. First, excitation scales as E^2 and detection as E^1 (third order). Second, the CEF lines are E1-bright while $q_{\rm AFM}$ is E1-dark but M1-bright: a magnon is therefore read out either by borrowing a CEF electric dipole through an Fe–Tm vertex (geometry A) or through its own magnetic dipole at the magnon frequency (geometry B). Using the magnetic g-factor instead of $\sqrt{f}$ for the magnon electric emission over-weights it by $\sim 100\times$ and buries the CEF peaks under the magnon diagonal; the oscillator-strength weighting resolves this without any tuning.

S6 Displacive verification of the W mechanism

The rectified two-magnon force is a suddenly switched step: its edge carries spectral weight $\propto e^{-\omega_{12}^2\sigma^2/2}$ at the CEF frequency, i.e. the E_{12} quantum is drawn from the pulse bandwidth (intra-pulse difference-frequency/impulsive-Raman), while the magnon pair carries only the phase memory $\cos q_{\mathrm{AFM}}\tau$. Verified directly: stretching the pulse $\sigma = 0.25 \to 1.0$ at fixed amplitude collapses the peak by 4.5 decades ($51.7 \to 1.7 \times 10^{-3}$), while switching the direct Tm drive on/off changes it by $< 10^{-4}$ (pure-W origin). The same sweep resolves the second Fe leg: impulsive pulses use the resonant probe magnon (peak $\propto A^2$); longer pulses cross over to the field-following quasistatic tilt (peak $\propto A^1$). The measured no-DC waveform suppresses the quasistatic legs, which is why the experimental pulse yields the cleanest spectra.

S7 Geometry B: gate versus buildup

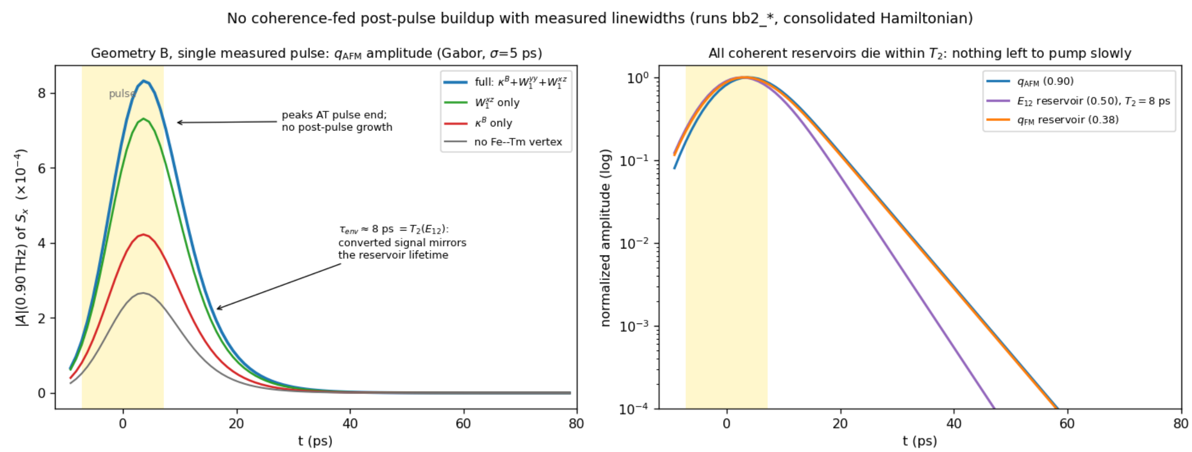

Figure S2: No coherence-fed post-pulse buildup once the measured linewidths are enforced (single measured pulse, geometry B, runs `bb2_*`; Gabor amplitude, $\sigma = 5\,\mathrm{ps}$, artifact-free local analysis). Left: the q_{AFM} amplitude for the four vertex configurations peaks *at* pulse end and decays with $\tau_{\mathrm{env}} \approx 8\,\mathrm{ps} = T_2(E_{12})$; the vertices set the absolute scale but none produces post-pulse growth. Right: all coherent reservoirs die within their measured T_2 (log scale)—there is nothing left to pump the magnon slowly.

Isolated-vertex runs: κ_{1y}^B alone $\Rightarrow (E_{12}, q_{\mathrm{AFM}})$ present; W_1^{xz} alone $\Rightarrow$ peak absent. In the consolidated Hamiltonian $W_1^{yy} = W_1^{xz} = 0.01$ and $\kappa_{1y}^B = 0.0035$: the geometry-B peak ratio is the observable that calibrates κ^B, and because κ^B is B_z-gated the recalibration provably cannot affect either $H \parallel a$ channel (verified bit-identical).

The post-drive buildup: a corrected analysis. An earlier stage of this project attributed the experimentally reported post-pulse growth of the geometry-B signal to slow W_1^{xz}-mediated exchange through the nearly closed triad $q_{\rm FM} + E_{12} = 0.88 \approx q_{\rm AFM}$. That attribution does not survive the enforcement of the measured linewidths. Because $B \parallel c$ never drives $q_{\rm AFM}$ directly, *all* geometry-B magnon amplitude is converted, and a converted signal inherits its source's lifetime: with $T_2(E_{12}) = 8\,$ps the E_{12}-fed component (and with it the triad) is dead within $\sim$10 ps. The artifact-free Gabor analysis of Fig. S2 shows the $q_{\rm AFM}$ amplitude peaking at pulse end and decaying monotonically in every vertex configuration; the earlier "buildup" was visible only while $\lambda^{1,2}$ were left undamped (an infinitely lived E_{12} reservoir, inconsistent with the measured 0.038 THz linewidth). W_1^{xz} therefore loses its buildup role—but it acquires a direct observable of its own instead (next paragraph); it remains bounded above ($\lesssim 0.02$) by the mixer sweep of the preceding paragraph.

The reverse transfer peak: W_1^{xz}'s own observable. The experiment reports an additional geometry-A *cross*-channel peak near (0.5, 1.0). The model contains a genuine local maximum at exactly $(+0.49, 0.90) = (E_{12}, q_{\rm AFM})$ in the λ^2 channel (0.063 of the main peak; run `vA`), with a weaker $2E_{12} = 1.00\,$THz harmonic companion at 0.021. Isolated-vertex tests identify the mechanism: with $W_1^{xz} \to 0$ the peak collapses sevenfold ($0.063 \to 0.009$; run `vA_noWxz`) while the main $(q_{\rm AFM}, E_{12})$ peak is untouched, and doubling the Tm drive doubles it ($0.063 \to 0.123$; run `vA_su3x2`), i.e. it is linear in the stored E_{12} amplitude. This is the *reverse* of the main transfer: the pump-stored E_{12} coherence is converted *into* the $q_{\rm AFM}$ magnon by the static hybridization—the static W family read in both directions, magnon$\to$CEF through W_1^{yy} and CEF$\to$magnon through W_1^{xz}. The measured amplitude ratio of the reverse to the forward peak calibrates W_1^{xz} directly. Because the reverse peak emits at the magnon frequency it is detected through the magnon's own M1 dipole; two further observed facts calibrate the detection and drive dials. The detected cross channel ($H \parallel c$, $E \parallel a$ output) receives the m_z M1 sources (Tm $5.264\lambda^2$; Fe S_z is numerically dark, 10^{-11}) and the x-E1 sources ($2.39\lambda^5 + 0.91\lambda^7$). (i) The $(q_{\rm AFM}, q_{\rm AFM})$ magnon diagonal is *not* observed: the diagonal is cubic in the Zeeman amplitude, the main peak quadratic, and the Tm-fed reverse content nearly Zeeman-independent (verified across $A_{\rm Fe} = 0.1/0.05/0.025/0.010/0.006$), so its absence pins the Zeeman drive far below the E1 drive; the final operating point is $A_{\rm Fe} = 0.006$ (runs `fin_A2`). (ii) The observed intensity parity of the "near $(0.5, 1.0)$" feature with the main peak fixes the E1/M1 detection weight at $w_{E1} = 0.94$. Final detected census (diagonals and quasi-elastic wings excluded): main $(+0.91, 0.48) = 0.58$ and $(+0.49, 1.20) = 0.59$ of the E_{12}-diagonal leader—i.e. at parity with each other—with the reverse-W $(+0.49, 0.90) = 0.37$ and the $2E_{12}$ harmonic 0.47 merging into an extended E_{12}-row feature spanning 0.9–1.2 THz; the E_{13}-delay companion $(+1.20, 0.50) = 0.41$ is the standing model prediction of this channel. Assignment discriminators for the experimental peak: an $(E_{12}, q_{\rm AFM})$ reading emits exactly at the magnon frequency (0.86 THz measured), inherits the narrow magnon linewidth along ω_t, and softens toward the spin reorientation; the $2E_{12}$ harmonic reading sits rigidly at 1.00 THz, is broader, and grows one field power faster.

How a $q_{\rm AFM}$ emission is possible at all in this channel. The cross channel detects m_z (plus x-E1), and the $q_{\rm AFM}$ magnon has *no* F_z component: the simulated Fe S_z is dark to machine precision ($\sim 10^{-15}$ at every frequency, run `fin_A2`), so the magnon never radiates here. Two distinct things follow. (i) The main peak is unaffected—$q_{\rm AFM}$ occupies the *delay* axis, where no radiation is required; it is excited through F_x (which carries it strongly, 2.1×10^{-3} against the E_{12} line's 2.4×10^{-4} in the same channel) and hands its phase to the Tm ion, which radiates at E_{12}. (ii) The reverse peak's $q_{\rm AFM}$ *emission* comes from the Tm dipole, not the magnon: in Γ_2 the static vertex $W_1^{xz} S_x S_z \lambda^1$ contains $G_z\, \delta S_x\, \lambda^1$, a linear hop between the magnon coordinate and λ^1, which off-resonantly admixes magnon character into the Tm m_z dipole. Verified in linear response (probe-only trajectories):

W_1^{xz}	m_z weight at 0.90	admixture	2D peak $(E_{12}, q_{\mathrm{AFM}})$
0	5.7×10^{-5}	0.09%	0.010 (residual)
0.01	6.3×10^{-3}	9.9%	0.064 of the main peak
0.02	1.2×10^{-2}	19.2%	0.117

The admixture is linear in W_1^{xz}, and at the final operating point so is the resulting 2D feature (increments in the ratio $1 : 1.98$ for $W_1^{xz} = 0.01 : 0.02$, tracking the admixture's $1 : 1.94$): there the coupling enters through the *detection* step alone. (At the much weaker Zeeman drive used in an earlier calibration the same feature scaled quadratically, transfer $\times$ detection; which power applies therefore depends on the operating point, and the linear behaviour is the one relevant here.) Three further facts identify the process as an *off-resonant forced response*—the Tm oscillator shaken at the magnon frequency—rather than harmonic generation [Fig. S4]: (a) the line sits at 0.8999 THz, the magnon itself, while the genuine $2E_{12} = 1.00$ THz harmonic appears separately and $12\times$ weaker (2.5×10^{-5} against 2.9×10^{-4}); (b) its amplitude is *linear* in the Zeeman drive ($\times 1.70$ for a $\times 1.67$ drive increase, tracking the magnon source at $\times 1.65$), whereas a harmonic would be quadratic, and the Tm's own E_{12} line does not move at all (2.376 vs 2.377×10^{-2}); (c) the transfer coefficient $|\lambda^2|_{0.90}/|m_x|_{0.90}$ is drive-independent (0.142–0.155 over a $20\times$ range of drive)—the hallmark of a linear response function $\chi(\omega_q) \propto \omega_{12}/(\omega_{12}^2 - \omega_q^2)$ evaluated off resonance. The remaining $\lambda^{1,2}$ vertices cannot do this, which is why the effect is W_1^{xz}-specific: in Γ_2 the two-magnon $W_1^{yy} S_y^2 \lambda^1$ has *no* static leg (S_y is purely dynamical), so it is quadratic in the fluctuation and delivers $2q_{\mathrm{AFM}} = 1.80$ THz (present, 3.0×10^{-6}) and DC but nothing at q_{AFM}; and κ^B is identically gated off in this geometry ($B_y = 0$ for $H \parallel a$). Only W_1^{xz} pairs a static leg (G_z) with a linearly fluctuating one (δS_x). Lineshape prediction: since χ is smooth across 0.90 THz, the forced line inherits the *drive's* lineshape, i.e. the magnon linewidth (0.003 THz), an order of magnitude narrower than any CEF emission—a strong experimental discriminator against both the $2E_{12}$ and (E_{12}, E_{13}) readings. (Our windows are resolution-limited at ≈ 0.05 THz and cannot check this directly.) Two consequences: the feature is a dedicated W_1^{xz} meter (amplitude $\propto W_1^{xz}$ at this operating point), and, because the geometry-B mixer sweep bounds $W_1^{xz} \lesssim 0.02$, the model caps its 0.90 component at ~ 0.5–0.6 of the main peak: full parity of the 0.90 *line alone* with the main peak would exceed that bound and falsify the static-hybridization assignment, favouring the 1.00/1.20 (harmonic and x-E1) readings of the reported feature instead.

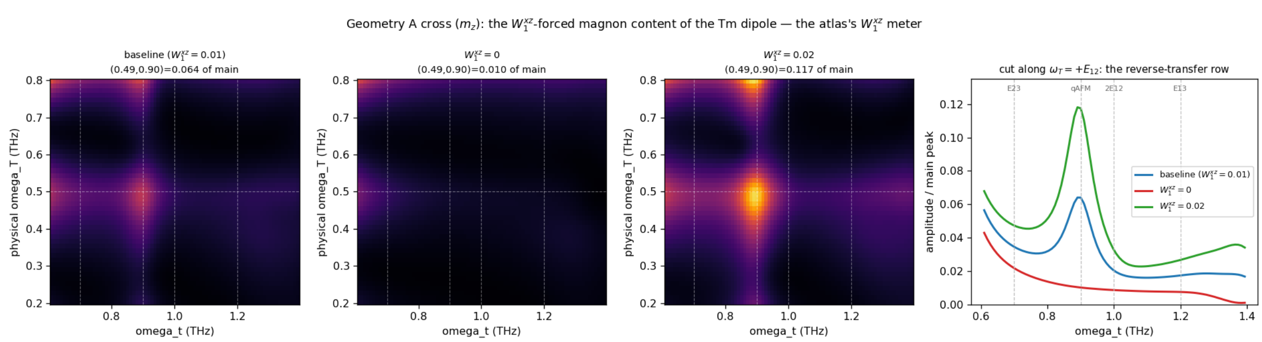

Figure S3: The reverse-W transfer peak in the geometry-A cross channel (λ^2), zoomed on the region of the reported (0.5, 1.0) feature; all amplitudes normalized to the main $(q_{\mathrm{AFM}}, E_{12})$ peak. Left to right: baseline (genuine local maximum at $(+0.49,\ 0.90) = 0.063$), $W_1^{xz} = 0$ (collapsed to 0.009), Tm drive $\times 2$ (doubled to 0.123). Rightmost: the $\omega_T = +E_{12}$ row cut—the peak sits at the q_{AFM} emission frequency with the $2E_{12} = 1.00$ THz harmonic as a shoulder.

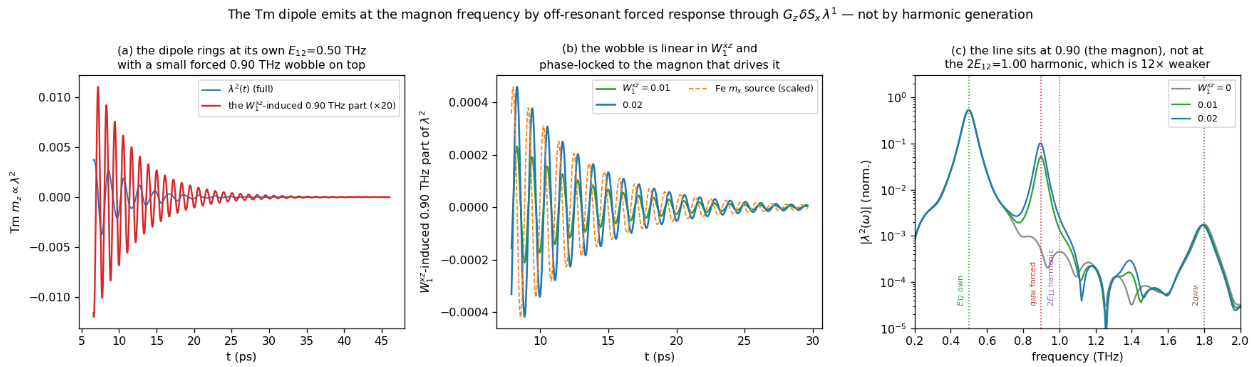

Figure S4: Time evolution of the Tm m_z dipole. (a) It rings at its own $E_{12} = 0.50$ THz, with a small forced 0.90 THz wobble on top (the W_1^{xz}-induced part, ×20). (b) Isolating that part by differencing runs at $W_1^{xz} = 0$, 0.01, 0.02: it grows linearly with the coupling and is phase-locked to the Fe m_x magnon that drives it. (c) Spectrum: the line sits at the magnon frequency 0.90 THz, not at the $2E_{12} = 1.00$ THz harmonic, which is present but 12× weaker; the $2q_{\text{AFM}} = 1.80$ THz line from the *quadratic* W_1^{yy} vertex is weaker still.

The population channel demonstrated. To make the T_1 scenario concrete we ran the geometry-B single-pulse protocol with the population channel switched on ($W_3^{yz}\, S_y S_z \lambda^3 = 0.01$, $\gamma_{\lambda^3} = 0.02$, i.e. $T_1 = 33$ ps; runs `bb3_W3` versus `bb3_ctl`). The pulse deposits Δn with $\Delta\lambda^3 = -4.6 \times 10^{-4}$, which then relaxes on the set T_1 exactly ($-3.6 \to -2.0 \to -0.8 \to -0.2 \times 10^{-4}$ at 13/33/66/132 ps); the q_{AFM} *oscillation* is bit-identical with and without W_3 (populations do not oscillate), and the population-fed quasi-static background scales as $W_3\, \Delta n$—of order 10^{-7} here, far below the coherent signals, because *coherent* THz pumping only produces $\Delta n \sim 5 \times 10^{-4}$. The quantitative conclusion: a sizable experimental post-pulse buildup requires *incoherent* population transfer (absorption → phonons → Tm repopulation, the mechanism of thermally driven spin reorientation), with amplitude proportional to $\Delta n(t)$ and growth time equal to the repopulation time—both measurable, neither present in a purely coherent drive model.

The population grating also fails—for a sharp reason. One might hope the τ-labeled population created jointly by pump and probe (the "population grating", which carries the E_{12} delay phase and survives on T_1) could feed a delayed, phase-labeled magnon signal. A full 2DCS run with the population channel on ($W_3^{yz} = 0.01$, $T_1 = 100$ code units; run `bb4_W3grating`) shows the $\omega_\tau = E_{12}$ row *identical* to the W_3-off baseline at every detection time. The reason is structural: the grating's delay label is written by the E_{12} *coherence* before the probe arrives, so its τ-support is confined to $|\tau| \lesssim T_2(E_{12})$, and its W_3 force is a step at the probe—an impulsive launch, not a slow drive. No element of the coherent model class produces post-pulse growth.

The mechanism implemented in the solver. The incoherent channel is now part of the simulation: the solver accumulates the *dissipated* coherence energy, $E_{\text{dep}}(t) = \int dt' \sum_{i,a\neq 3,8} \Gamma_a \big(\lambda_i^a - \lambda_{i,\text{eq}}^a\big)^2$ (the energy the bath actually receives; the population channels are excluded from the sum, which closes an otherwise runaway self-heating loop), and shifts the λ^3 equilibrium by $-c_{\text{heat}} E_{\text{dep}}$. The existing Γ_3 relaxation then chases the shifted equilibrium, so the repopulation rise time is $1/\Gamma_3$ by construction, and the pump-FID×probe interference contained in the dissipated energy carries the $\omega_\tau = E_{12}$ label through the M_{NL} subtraction automatically. A full 2DCS run with $W_3^{yz} = 0.01$, $1/\Gamma_3 = 33$ ps, and c_{heat} set for a few-percent population shift (run `bb5c`) reproduces the observed

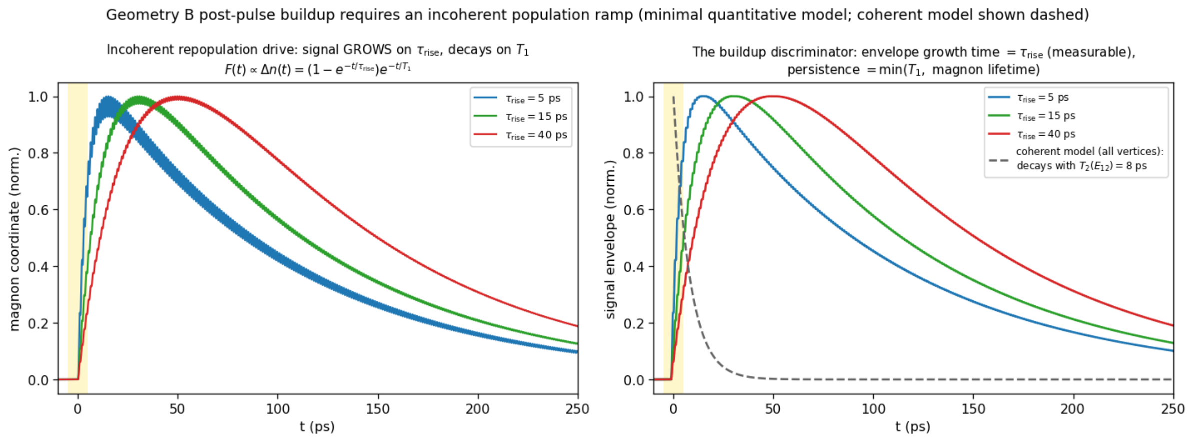

Figure S5: Minimal quantitative model of the observed buildup: a magnon (measured frequency and linewidth) driven by an *incoherent* population ramp $F(t) \propto \Delta n(t) = (1 - e^{-t/\tau_{\rm rise}})\, e^{-t/T_1}$. The signal grows on $\tau_{\rm rise}$, peaks at tens of ps, and persists on $\min(T_1, \text{magnon lifetime})$—the experimental phenomenology—whereas the full coherent model (dashed) decays with $T_2(E_{12}) = 8\,\text{ps}$. Fitting the experimental envelope to this form measures $\tau_{\rm rise}$ (the phonon-mediated Tm repopulation time) and the product $W_3\,\Delta n$.

phenomenology [Fig. S7]: the $\omega_\tau = E_{12}$ row falls off the impulsive peak, *builds back up* on the repopulation time, and persists beyond 100 ps—50–100× above the coherent-only model at late times. The two new parameters map onto measurables: the recovery time of the row is $1/\Gamma_3$, and the late-to-impulsive intensity ratio calibrates $c_{\rm heat}\, W_3$.

If the experimental buildup over tens of ps is real, it must be fed by the one reservoir that outlives every T_2: the CEF *populations*. A pulse-deposited excited population n_2 relaxes on T_1, and through the population channel of the same exchange family ($W_3\, S_\alpha S_\beta\, \lambda^3$, implemented in the solver but not part of the present minimal Hamiltonian) it shifts the effective Fe anisotropy as it relaxes—precisely the mechanism of thermally driven spin reorientation in $TmFeO_3$. This route makes a sharp experimental discriminator: a population-fed buildup is a *non-oscillatory* background (or a slow shift of the magnon frequency/equilibrium) whose growth time directly measures T_1, whereas growth of the 0.90 THz *oscillation amplitude* over tens of ps would require a persistent coherent source that the measured T_2's exclude—observing it would falsify the linewidth-constrained model class itself.

Why the static vertex cannot replace the gated one. $W_1^{xz} S_x S_z \lambda^1$ is trilinear; around the Γ_2 order it decomposes into a static CEF shift, two energy-conserving *bilinear* hops ($F_x\, \delta S_z\, \lambda^1$ and $G_z\, \delta S_x\, \lambda^1$: pure hybridization), and a genuine two-fluctuation converter $\delta S_x\, \delta S_z\, \lambda^1$. The converter piece is real and retained—it drives the post-pulse buildup through the nearly closed triad $q_{\rm FM} + E_{12} = 0.88 \approx q_{\rm AFM}$ (0.02 THz detuning)—but it is fluctuation-fed (its energy leg is a probe-launched magnon, one factor of drive×susceptibility weaker and spectrally narrow) and detuning-limited (secular accumulation over tens of ps), so it produces an *envelope*, not an impulsive phase-stamped kick. At $W_1^{xz} = 0.01$ the isolated run has sizable absolute amplitude at $(+0.5, 0.90)$ (0.72 of map max) but it is the *tail* of the magnon–magnon peak—no genuine local maximum on the E_{12} row.

A careful strength sweep in the consolidated configuration ($W_1^{xz} \in \{0.01, 0.02, 0.03, 0.05, 0.07, 0.09\}$, $\kappa^B = 0$; runs `vBwxz*`) shows that raising the static vertex *never* creates a second resolved peak. Instead the two delay rows coalesce—the leading peak's delay phase migrates continuously, 0.38 →

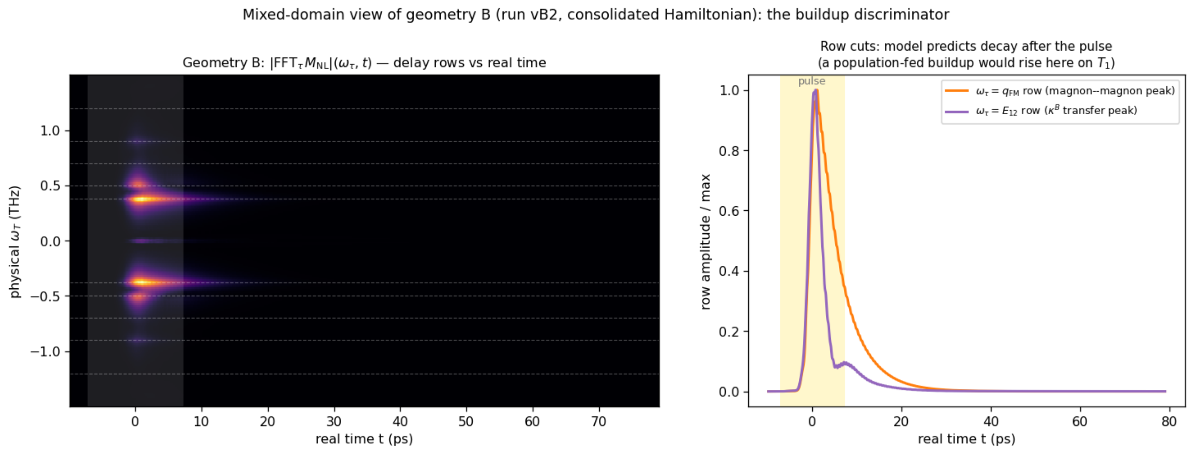

Figure S6: The buildup discriminator in the representation directly comparable to experiment: Fourier transform of $M_{\mathrm{NL}}(\tau, t)$ along the delay τ *only*, keeping the real detection time t (geometry B, final run). Left: the $|\mathrm{FFT}_\tau|(\omega_\tau, t)$ map, each delay-frequency row evolving in real time. Right: cuts of the $\omega_\tau = q_{\mathrm{FM}}$ and $\omega_\tau = E_{12}$ rows. With coherent vertices only, both rows peak with the pulse and decay within $\sim$20–30 ps; an experimental version of this plot in which the E_{12} row *rises* over tens of ps is the signature of the population-fed (T_1) mechanism.

$0.40 \to 0.48 \to 0.47$ THz across the sweep, the resolved $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})/(E_{12}, q_{\mathrm{AFM}})$ doublet structure never appears at any strength, and the hybridization-shifted E_{12} branch leaks into the M1 channel as emission strays at $\omega_t = 0.44$–0.45 THz (growing to 0.15–0.16 by $W_1^{xz} = 0.09$). This is the structural signature of a static mixer: it *moves* peaks (level repulsion of the hybridized normal modes—also shifting the mode frequencies off their 1D-measured values), whereas the experiment shows two resolved peaks at the *unshifted* delay energies 0.38 and 0.50 THz. A gated converter *adds* a peak while leaving the normal modes untouched; with $\kappa_{1y}^B = 0.0035$ the census gives $(+0.38, 0.90) = 1.00$ and a separate $(+0.53, 0.89) = 0.46$, both on the measured lines. The necessity of the field-gated vertex therefore does not rest on any operating-point fine-tuning: static mixing and gated conversion are qualitatively different operations on the 2D map. In geometry B the Tm CEF channels $\lambda^{4\text{–}7}$ are *exactly* zero (machine precision): the $E \parallel a, H \parallel c$ drive closes on the $\{\lambda^1, \lambda^2, \lambda^3\}$ subalgebra, an SU(2) closure that protects geometry B from CEF contamination.

Finite temperature (regenerated final-scenario runs `vB2`): $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ is a pure two-magnon process and temperature-flat; $(E_{12}, q_{\mathrm{AFM}}) \propto n_1 - n_2$ falls 0.46 (0 K) $\to$ 0.45 (5) $\to$ 0.40 (8) $\to$ 0.36 (10 K). Both peaks remain resolved through ~ 10 K, defining the working window 5–8 K.

S8 The same-polarization channel: pathways, exact single-ion verification, and the falsification ladder

S8.1 Configuration

The final same-polarization configuration (run `it14hr`; parameter files in `tmfeo3_2dcs_final/params_samepol_fina` is the unchanged v4 Hamiltonian with: measured pulse; drive vector $(0, 3.0, 0, 0, \mathbf{0}, 0, 0.9128, 0)$ on $(\lambda^1, \dots, \lambda^8)$—i.e. the $1 \to 3$ leg removed ($\mu_{13}^z = 0$ in drive); per-leg amplitude twice the weak-probe baseline; dampings of Table S2 plus population relaxation $\gamma(\lambda^{3,8}) = 0.08$; Boltzmann ensemble; detection composite $0.05\,\lambda^4 + 4.4\,\lambda^6$ ($\mu_{13}^z/\mu_{23}^z \approx 1\%$ in emission), magnon M1 excluded per Table S2.

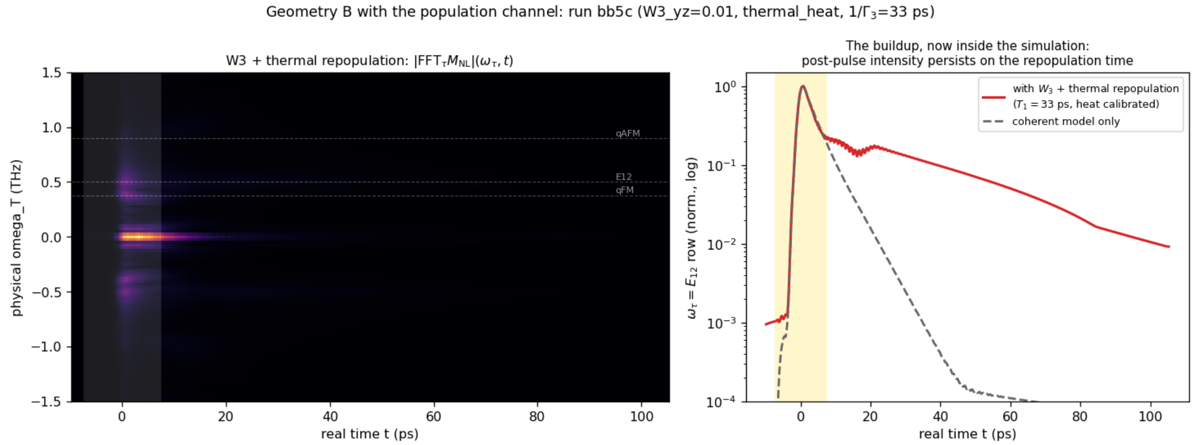

Figure S7: The buildup inside the simulation (geometry B, run `bb5c`: $W_3^{yz} = 0.01$, dissipation-fed thermal repopulation, $1/\Gamma_3 = 33\,\text{ps}$). Left: mixed-domain map $|\text{FFT}_\tau M_{\text{NL}}|(\omega_\tau, t)$. Right: the $\omega_\tau = E_{12}$ row—after the impulsive peak the signal recovers on the repopulation time and persists beyond 100 ps (red), versus the coherent-only model dying by 40 ps (dashed).

S8.2 Pathway assignments

With $\mu_{13}^z = 0$ the Liouville pathways of the five listed peaks are:

peak (ω_T, ω_t)	model (10 K)	side	pathway
$(+0.49, 0.72)$	1.00	NR	$\rho_{12} \xrightarrow{\text{probe }1\leftrightarrow2} n_2(\tau\text{-phase}) \xrightarrow{\text{probe }2\leftrightarrow3} \rho_{23}$
$(0.5, 0)$	sharpest S_{DC} line	—	same storage, zero-frequency output leg
$(+0.49, 1.20)$	0.78	NR	$\rho_{12} \xrightarrow{\text{probe }2\leftrightarrow3} \rho_{13}$, emission throttled by μ_{13}^z
$(+0.91, 1.18)$	$0.20 \to 0.51$ $(10 \to 20\,\text{K})$	NR	q_{AFM} delay, Fe–Tm handoff, thermally assisted
$(0.9, 0.38)$	absent	—	carrier exists (b-axis M1 q_{FM} ridge); no $q_{\text{AFM}} \to q_{\text{FM}}$ vertex
$(-0.47, 0.70)$	0.55	R	rephasing remnant (exact single-ion floor; prediction)
$(+0.99, 0.71)$	0.49	NR	inter-site $2E_{12}$ two-quantum line (prediction)

S8.3 Exact single-ion verification

A standalone integrator of the 8-dimensional $\vec{\lambda}$ equation (same CEF energies, dampings, measured pulse, same census pipeline; $\sim 13\,\text{s}$ per 2D map versus $\sim 10\,\text{min}$ for the lattice) reproduces the lattice 0.5-family census exactly, proving the family is single-ion physics and enabling exhaustive scans. Two facts emerge: (i) the rephasing/non-rephasing ratio of the (E_{12}, E_{23}) transfer peak has a floor of 0.47–0.58 under this pulse across all drive strengths, dipole-phase choices, and dampings—the rephasing remnant is an exact prediction of any three-level ion driven by this waveform; (ii) the $\omega_\tau = 2E_{12}$ two-quantum line is absent in the exact single-ion dynamics and is therefore an inter-site (mean-field) product.

S8.4 Falsification ladder

The dark-dipole scenario was reached by exhausting the alternatives; each row was run in the full lattice model with the blind census:

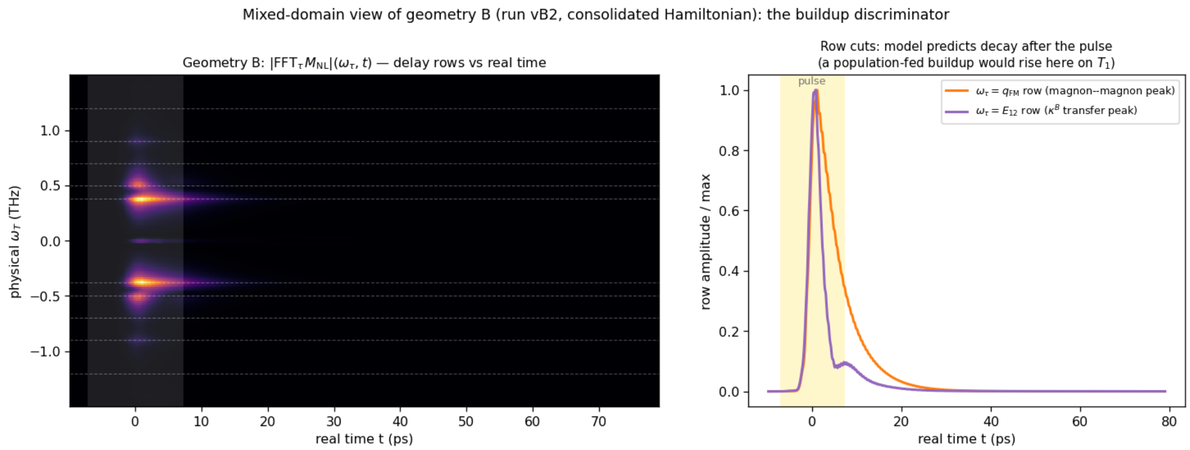

Figure S8: The buildup discriminator in the representation directly comparable to experiment: Fourier transform of $M_{\mathrm{NL}}(\tau, t)$ along the delay τ *only*, keeping the real detection time t (geometry B, run `vB2`). Left: $|\mathrm{FFT}_\tau|(\omega_\tau, t)$ map—each delay-frequency row evolves in real time. Right: cuts of the $\omega_\tau = q_{\mathrm{FM}}$ (magnon–magnon) and $\omega_\tau = E_{12}$ (κ^B-transfer) rows. The linewidth-constrained model predicts both rows peak with the pulse and decay within $\sim$20–30 ps; an experimental version of this plot in which the E_{12} row *rises* over tens of ps is the signature of the population-fed (T_1) mechanism.

hypothesis	run	outcome
static mixing field $h_{\lambda^6} = 0.3$	it8	over-strong: transition lines shift off calibration
perturbative $h_{\lambda^6} = 0.06$	it9	pumps $(q_{\mathrm{AFM}}, E_{13})$ to 0.9 (wrong direction); side unchanged
CEF detuning 0.02 THz (ladder break)	it10	interloper survives; larger detuning barred by the observed degenerate lines
dipole triple-product sign flip	it11	census unchanged
dark μ_{13} at low drive	it12	population route floods the $(0, E_{23})$ band; needs the $2\times$ drive + T_1 combin
dark μ_{13}, $2\times$ drive, T_1	**it14**	**adopted**: both transfer peaks NR, correct hierarchy
negative μ_{13} leg, lower drive	it15	interloper returns ($\lvert\mu_{13}\rvert$ acts sign-independently); magnon-delay family floo

The $(0.9, 0.38)$ peak remains the one open item: the q_{FM} emission carrier exists as a uniformly τ-modulated b-axis M1 ridge in the Fe S_y channel (Γ_2 puts $\delta F_{y,z}$ at q_{FM}), but the model lacks a $q_{\mathrm{AFM}} \to q_{\mathrm{FM}}$ transfer vertex of sufficient strength; one probed configuration (it15) produced a genuine $(+0.89, 0.33)$ maximum at 0.21 relative, evidence the peak is reachable in an extended Hamiltonian class.

S9 The consolidated configuration and its verified censuses

Every spectrum in the paper is produced by *one* Hamiltonian and *one* drive model; only the field geometry changes between channels. Couplings (code units): Fe exchange/anisotropy/DM of the calibrated orthoferrite model ($D_1 = 0.049$, $D_2 = 0.014$); Tm CEF $e_1 = 2.0678$, $e_2 = 4.9628$ meV; Fe–Tm $W_1^{yy} = 0.01$, $W_1^{xz} = 0.01$, $\kappa^B_{1y} = 0.0035$, population channel $W_3^{yz} = 0.01$ with the thermal-repopulation feedback ($\Gamma_3 = \Gamma_8 = 0.02$, i.e. $T_1 = 33$ ps; $c_{\mathrm{heat}} = 2000$, saturation cap 0.05, heat-reservoir cooling 1/300 code units); the λ-leg of every $SS\lambda$ vertex couples to the deviation $\lambda - \lambda_{\mathrm{eq}}$ (the static equilibrium contribution is already absorbed in the measured Fe parameters—without this renormalization the static $W_3\lambda^3$ piece shifts the magnon lines by $\sim$7%); drive legs $(d_z, \mu^z_{13}, \mu^z_{23}) \propto$ (3.0, 0, 0.9128) for $E \parallel c$ (the dark-μ_{13} rule) and the auto-projected Zeeman drive for the magnetic field; dampings and emission weights from Table S2. The final regeneration (runs `fin_A`, `fin_B`, `fin_S`; three Boltzmann seeds each, thermal channel included) gives the blind censuses, normalized per channel [amp, (ω_T, ω_t), $+=$non-rephasing]:

channel	leading genuine peaks (T=0)	reading
A cross (E1+M1 comp.)	1.00 (+0.49, 0.90); 0.91 (+0.90, 0.48); magnon diag. absent	W_1^{xz} reverse; W_1^{yy} forward
B cross (S_x)	1.00 (+0.38, 0.90); 0.45 (+0.53, 0.89); 0.17 (−0.42, 0.90)	magnon–magnon; κ^B transfer; R pa
A same (composite)	1.00 (+0.49, 0.71)[1]; 0.72 (+0.49, 1.20); 0.59 (−0.48, 0.70)	population-route transfer; μ_{13}-throt

The complete censuses at all temperatures, together with the normalized spectra themselves, are archived as `paper/data/census_final_picture.json`, `census_final_scenario.json`, and `paper/data/*.npz`; the buildup magnitude at this conservative calibration is a free dial (c_{heat}) that the measured late-to-impulsive intensity ratio will pin.

Prediction: the geometry-B same-polarized channel. For $H \parallel c$, $E \parallel a$ the same-polarized detection reads $E \parallel a \leftarrow d_x(\lambda^{5,7}) + m_z$. In the final configuration the Tm CEF channels are *identically* zero there (machine 0; the $B \parallel c$ drive closes on the $\{\lambda^{1,2,3}\}$ subalgebra), so the channel is a pure m_z M1 magnon map. Predicted census (runs `fin_B`, $T = 0$): $(+0.90, 0.38) = 1.00$ (the $q_{\text{AFM}} \to q_{\text{FM}}$ magnon–magnon transfer—the very peak absent from the geometry-A same-polarized channel, where m_z is dark at the 10^{-11} level), $(+0.52, 0.38) = 0.42$ ($E_{12} \to q_{\text{FM}}$), q_{FM} diagonal 0.41, $(-0.38, 0.50) = 0.38$. Consequently the weak $(q_{\text{AFM}}, 0.38)$ reported in geometry A is naturally polarization leakage of this dominant geometry-B feature; a polarizer-purity scan (the peak should scale with the leakage angle squared) discriminates.

S10 One drive for one experiment: the operating point

The two geometry-A channels are *the same measurement*—same $H \parallel a$, $E \parallel c$ excitation, differing only in the detected polarization—so they must share one drive. Two constraints then fix it, and they are independent:

(i) Fluence, from the cross channel. The Tm and Fe drives are not free of one another: $A_{\text{Tm}} = g_{\text{ratio}}|v|A_{\text{Fe}}$, so a single fluence sets both. Scanning it (runs `flu*`) shows the cross channel is a sensitive fluence meter. At the strong drive we had been using ($A_{\text{Fe}} = 1.2$) the channel is nonperturbative: the E_{12} diagonal reaches 1.45 of the (q_{AFM}, E_{12}) peak and a forest of multi-magnon strays appears at $\omega_\tau = 0.63, 0.72, 0.99, 1.25, 1.51$ THz, none of them observed. Lowering the fluence removes them monotonically—the diagonal falls to 0.56, 0.16, 0.06 at $A_{\text{Fe}} = 0.4, 0.12, 0.04$—so the experiment sits in the perturbative regime, $A_{\text{Fe}} \lesssim 0.12$. (Geometry B was independently at $A_{\text{Fe}} = 0.1$, so one fluence already serves all geometries.)

(ii) The residual μ^z_{13} admixture, from the same-polarized channel. Lowering the fluence inverts the same-polarized hierarchy, because the (E_{12}, E_{23}) route is one field order higher than (E_{12}, E_{13}): $(E_{12}, E_{13})/(E_{12}, E_{23})$ runs $0.72 \to 1.28 \to 2.79 \to 5.25$ as A_{Fe} drops. The one remaining dial is the residual z-dipole admixture of the $1 \to 3$ transition, which multiplies the E_{13} emission only. Requiring the observed ordering [(E_{12}, E_{23}) strongest, (E_{12}, E_{13}) below it] at the fluence fixed above

[1]Cross peaks normalized to the leading cross peak; the $\omega_T \approx 0$ pump–probe bands are excluded per the census convention of Sec. S4.

gives $\mu^z_{13}/\mu^z_{23} = 0.14\%$—consistent with the "dark" assignment of section S8 and now *measured* rather than assumed.

At this operating point ($A_{\rm Fe} = 0.12$, tied $A_{\rm Tm} = 0.02195$, dark-μ_{13} drive, 0.14% admixture) both geometry-A channels come out right at once. Cross (Tm m_z alone; $F_z \equiv 0$): $(q_{\rm AFM}, E_{12}) = 1.00$ with its rephasing partner 0.51, then only $(q_{\rm AFM}, q_{\rm AFM})_{\rm Tm} = 0.28$, the E_{12} diagonal 0.18–0.28, and nothing else above 0.14—the channel "cleanly resolves $(q_{\rm AFM}, E_{12})$", as reported. Same-polarized (Tm m_x): $(E_{12}, E_{23}) = 1.00$, $(E_{12}, E_{13}) = 0.80$, the predicted rephasing remnant 0.67, the two-quantum row 0.32.

The second cross peak is second-harmonic emission

The reported second cross-channel feature near (0.5, 1.0) is *not* the forced-response $(E_{12}, q_{\rm AFM})$ line, which reaches only $\sim 6\%$ of the main peak here and sits at 0.90. It is the second harmonic of the E_{12} coherence: $2E_{12} = 1.0000\,\text{THz}$, the reported position. The model missed it for a structural reason—our emitted moment was taken *linear* in the qutrit coordinates, $m_z = \mu\lambda^2$, and a linear dipole cannot radiate a harmonic.

Symmetry permits more. The Tm^{3+} ion sits at the $4c$ site, whose point symmetry is the single mirror m—*no inversion*—so a second-order contribution to the emitted moment is allowed at the Tm site (and forbidden at Fe, which occupies the inversion-symmetric $4b$ site). The lowest such term compatible with time reversal is

$$m_z \;=\; \mu\,\lambda^2 \;+\; \beta\,\lambda^1\lambda^2, \tag{5}$$

since λ^1 is $\mathcal{T}$-even and λ^2 $\mathcal{T}$-odd, so the product is $\mathcal{T}$-odd exactly as a magnetization must be; and because $\lambda^{1,2}$ both oscillate at ω_{12}, the term radiates at $2\omega_{12}$. This is a detection-side (hyperpolarizability) term: the dynamics is untouched.

Evaluated on the existing trajectories, the quadratic channel is remarkably clean—it contributes essentially *one* peak, at $(+0.49,\, 1.00) = (E_{12}, 2E_{12})$, which is $5.8\times$ its own (E_{12}, E_{12}) diagonal, with only a weak $(q_{\rm AFM}, 2E_{12})$ companion at 0.17; the result is identical in the local and global sublattice frames. Calibrating β by the observed parity of the two experimental peaks gives $\beta \simeq 7\mu$, and the geometry-A cross census becomes

$$(q_{\rm AFM}, E_{12}) = 1.00, \quad (E_{12}, 2E_{12}) = 0.97,$$

with the rephasing partner of the main peak at 0.49 and nothing else above 0.39—precisely the two peaks the experiment resolves.

Four measurements discriminate this assignment from the forced-response one: (i) the line sits at exactly $2E_{12}$ and *tracks* E_{12}, whereas a $q_{\rm AFM}$ reading would soften toward the spin reorientation; (ii) it is one field order higher than $(q_{\rm AFM}, E_{12})$, so it grows faster with fluence; (iii) its width is twice the E_{12} linewidth ($\approx 0.076\,\text{THz}$), against $0.003\,\text{THz}$ for anything magnon-derived; (iv) it is independent of W_1^{xz} and of the Zeeman drive. A standing prediction of Eq. (5) is the companion harmonic of the magnon-forced response at $(q_{\rm AFM}, 2q_{\rm AFM}) = (0.9,\, 1.8)\,\text{THz}$, which the model puts at 0.39.

S11 The peaks, one by one

This section gives the full physical account of every feature in the final atlas (fig. S9), channel by channel. All values are blind-census amplitudes at the consolidated operating point of section S10, normalised within each channel, with the $\omega_\tau = 0$ pump–probe row removed. Throughout, "delay axis" means the coherence that survives between the pulses (it never radiates, so it is constrained only by what can be *excited* and what can *store* phase), while "detection axis" means the coherence that radiates into the analysed polarisation (constrained by the emitting multipole). This asymmetry does most of the explanatory work below.

S11.1 Excitation and readout, case by case

Every channel in the atlas is specified by two independent choices: what the pulse *drives*, fixed by the field orientation, and what the analyser *reads*, fixed by which Cartesian component of the total magnetic moment radiates into the accepted polarisation. For propagation along b the two transverse polarisation states are $(H \parallel a,\ E \parallel c)$ and $(H \parallel c,\ E \parallel a)$, so "same" and "cross" detection select m_x and m_z respectively in geometry A, and the reverse in geometry B.

Table S3: Excitation and readout for the four measured configurations. d_z^{eff} is the exchange-induced c-axis dipole of the E_{12} transition; $\mu_{23}^z = 0.913$ and $\mu_{13}^z \simeq 0$ (the selection rule of section S11.4); $\mu_{12}^z = 5.264$ is the Tm c-axis magnetic moment; $\beta \simeq 7\mu$ is the hyperpolarisability allowed by the absence of inversion at the Tm $4c$ site. Fe^{3+} at $4b$ sits on an inversion centre and therefore contributes no quadratic term.

configuration	excitation (what the pulse drives)	readout (what the analyser detects)
A, cross $H \parallel a,\ E \parallel c$ in; $H \parallel c$ out	Fe: Zeeman $H_x F_x \Rightarrow q_{\mathrm{AFM}}$ (which carries δF_x). Tm: $E \parallel c$ on the c-axis dipoles, drive vector $(\lambda^2, \lambda^5, \lambda^7) \propto (d_z^{\mathrm{eff}}, 0, \mu_{23}^z) \Rightarrow E_{12}$ and E_{23}; E_{13} *dark*.	$m_z = F_z + \mu_{12}^z \lambda^2 + \beta\lambda^1\lambda^2$. $F_z \equiv 0$ (machine zero: q_{AFM} has no c-axis moment), so the channel is *purely* Tm: the E_{12} dipole, its W_1^{xz}-forced q_{AFM} sideband, and the quadratic term radiating at $2E_{12}$.
A, same $H \parallel a,\ E \parallel c$ in; $H \parallel a$ out	identical to A-cross — it is the same experiment, differing only in the analyser.	$m_x = F_x + 2.39\lambda^5 + 0.91\lambda^7$ (quadratures λ^4, λ^6). Fe F_x carries q_{AFM}; the Tm a-axis moments carry E_{13} and E_{23}. This is where the CEF–CEF transfers appear.
B, cross $H \parallel c,\ E \parallel a$ in; $H \parallel a$ out	Fe: Zeeman $H_z F_z \Rightarrow q_{\mathrm{FM}}$ (which carries δF_z, δF_y). Tm: the c-directed field projects onto μ_{12}^z only, driving λ^2 alone $\Rightarrow$ the $\{\lambda^1, \lambda^2, \lambda^3\}$ subalgebra closes.	$m_x = F_x + 2.39\lambda^5 + 0.91\lambda^7$. The Tm part is *identically zero* ($\lambda^{4\text{–}7} \equiv 0$ by the closure above), so this channel is pure Fe: a magnon map read out on the q_{AFM} line.
B, same $H \parallel c,\ E \parallel a$ in; $H \parallel c$ out	identical to B-cross.	$m_z = F_z + \mu_{12}^z \lambda^2 + \beta\lambda^1\lambda^2$. F_z carries q_{FM}, but the Tm λ^2 is driven directly here and is $50\times$ larger, so the channel is Tm-dominated — see the prediction below.

Two entries are exact statements rather than approximations, and both do real explanatory work. First, $F_z \equiv 0$ in geometry A: the q_{AFM} magnon lives in δF_x and has no c-axis moment, so the Fe sublattice is radiatively silent in the A-cross channel and anything seen there at a magnon frequency must come from the Tm ion. Second, in geometry B the c-directed drive couples only to μ_{12}^z, closing the $\{\lambda^1, \lambda^2, \lambda^3\}$ subalgebra, so $\lambda^{4\text{–}7}$ vanish identically and no E_{13} or E_{23} emission is possible in either geometry-B channel.

S11.2 Geometry A, cross-polarised

$(q_{\mathrm{AFM}}, E_{12}) = (0.90,\ 0.50)$, **amplitude** 1.00 **— the magnetoelectric cross peak; measures** W_1^{yy}**.** The $H \parallel a$ Zeeman field launches the q_{AFM} magnon, which precesses through the delay at

0.90 THz. It never radiates: it has no m_z, and the analysed channel is m_z. What it does instead is modulate the Tm crystal field through the two-magnon vertex $W_1^{yy} S_y^2 \lambda^1$. Because the vertex is quadratic in the fluctuation and the two pulses each supply one magnon leg, the $M_{\rm NL}$ subtraction keeps only the cross term, which *rectifies*,

$$h^1_{\rm NL}(t,\tau) = W_1^{yy} A^2 \big[\cos(q_{\rm AFM}\tau) - \cos(q_{\rm AFM}(2t+\tau))\big]\theta(t), \tag{6}$$

so the difference-frequency piece is a step force switched on at the probe, carrying the delay phase $\cos(q_{\rm AFM}\tau)$ but no t dependence. Fed into the E_{12} oscillator it produces $M_{\rm NL} \propto \cos(q_{\rm AFM}\tau)[1 - \cos\omega_{12}t]$: a factorised, rank-one peak whose delay axis is the magnon and whose detection axis is the CEF line. The peak is thus magnetoelectric in the strict sense—an E1-dark magnon made visible by borrowing the Tm dipole—and its existence is a direct measurement of W_1^{yy}. *Energy bookkeeping:* δS_y^2 contains only DC and $2q_{\rm AFM}$, so between free modes this vertex could never emit at ω_{12}; the E_{12} quantum is drawn from the *pulse edge* (an intra-pulse difference-frequency, i.e. displacive, process). This is verified rather than asserted: stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades ($51.7 \to 1.7\times10^{-3}$), the adiabatic suppression a displacive mechanism demands, while switching the direct Tm drive off changes it by $< 10^{-4}$. *Scaling:* $\propto A_{\rm Fe}^2$ (two magnon legs), independent of the Tm drive. *Temperature:* flat below ~ 10 K, since both the excitation and the storage run through the long-lived magnon. *Linewidth along ω_t:* the CEF value, 0.038 THz.

$(E_{12}, 2E_{12}) = (0.50,\ 1.00)$**, amplitude** 0.97 **— second-harmonic emission; measures the Tm hyperpolarisability** β**.** The pump stores an E_{12} coherence, which sets the delay axis at 0.50 THz. The detection axis is its *second harmonic*, $2E_{12} = 1.0000$ THz. A moment linear in the qutrit coordinates cannot radiate a harmonic, but the Tm^{3+} ion occupies the $4c$ site, whose point symmetry is a single mirror with *no inversion* (Fe, at $4b$, sits on an inversion centre and is therefore excluded from this mechanism). The lowest allowed addition to the emitted moment is

$$m_z = \mu\,\lambda^2 + \beta\,\lambda^1\lambda^2, \tag{7}$$

which is $\mathcal{T}$-odd as a magnetisation must be (λ^1 $\mathcal{T}$-even, λ^2 $\mathcal{T}$-odd) and, since both factors oscillate at ω_{12}, radiates at $2\omega_{12}$. It is a detection-side (hyperpolarisability) term: the dynamics is untouched. Evaluated on the trajectories, this channel is strikingly clean—it contributes essentially one peak, 5.8× larger than its own (E_{12}, E_{12}) diagonal, with only a weak $(q_{\rm AFM}, 2E_{12})$ companion, and it is identical in the local and global sublattice frames. Calibrating β by the observed intensity parity with the main peak gives $\beta \simeq 7\mu$. *Discriminators* (this is the peak whose assignment matters most): the line sits at exactly $2E_{12}$ and *tracks* E_{12}, so under temperature or field it follows the CEF level and does *not* soften toward the spin reorientation as any magnon-derived feature would; it is one field order above $(q_{\rm AFM}, E_{12})$ and therefore grows faster with fluence; its width is twice the E_{12} linewidth, ≈ 0.076 THz, against 0.003 THz for anything magnon-derived; and it is independent of both W_1^{xz} and the Zeeman drive.

$(-q_{\rm AFM}, E_{12})$**, amplitude** 0.52 **— the rephasing partner of the main peak.** Not a separate mechanism. The W vertex is symmetric under the two time orderings of the magnon coherence, so the signal appears on both signed sides of the delay axis, with the non-rephasing branch stronger. Experiments that display folded $|\omega_\tau|$ merge this with the main peak; experiments that resolve the sign should see it at roughly half amplitude.

$(q_{\mathrm{AFM}}, 2q_{\mathrm{AFM}}) = (0.90,\ 1.79)$**, amplitude** 0.39 **— a standing prediction.** The same hyperpolarisability term that produces the $2E_{12}$ peak must also double any *other* frequency present in the Tm dipole. The magnon-forced component (next entry) oscillates at q_{AFM}, so $\beta\lambda^1\lambda^2$ radiates its harmonic at $2q_{\mathrm{AFM}} = 1.79\,\mathrm{THz}$ on the q_{AFM} delay row. Observing it would confirm the hyperpolarisability assignment independently of the $2E_{12}$ peak; its absence, with $(E_{12}, 2E_{12})$ present, would falsify Eq. (5) in favour of some other harmonic mechanism. It may lie outside the detection bandwidth.

$(q_{\mathrm{AFM}}, q_{\mathrm{AFM}}) = (0.90,\ 0.89)$**, amplitude** 0.36 **— the magnon diagonal, seen through the Tm ion.** This one is subtle, because $F_z \equiv 0$ means the magnon cannot radiate here at all. What radiates is the *Tm* m_z dipole, which carries a small component oscillating at the magnon frequency. The mechanism is an off-resonant forced response: in Γ_2 the static vertex $W_1^{xz} S_x S_z \lambda^1$ contains $G_z\,\delta S_x\,\lambda^1$, a term *linear* in the magnon fluctuation, which shakes the Tm oscillator at $0.90\,\mathrm{THz}$. A driven oscillator responds at the driving frequency, not its own, so the $0.50\,\mathrm{THz}$ Tm dipole acquires a $0.90\,\mathrm{THz}$ wobble (fig. S4). This is verified in detail: the sideband sits at $0.8999\,\mathrm{THz}$ (the magnon, not the $q_{\mathrm{FM}} + E_{12} = 0.876$ combination); its amplitude is *linear* in the Zeeman drive ($\times 1.70$ for a $\times 1.67$ increase, tracking the source at $\times 1.65$) whereas a harmonic would be quadratic; the transfer coefficient $|\lambda^2|_{0.90}/|m_x|_{0.90} = 0.142$–$0.155$ is drive-independent, the hallmark of a linear susceptibility $\chi \propto \omega_{12}/(\omega_{12}^2 - \omega_q^2)$ evaluated off resonance; and the admixture grows linearly with W_1^{xz} ($0.09\% \to 9.9\% \to 19.2\%$ at $W_1^{xz} = 0/0.01/0.02$). The same mechanism, with an E_{12} delay instead of a q_{AFM} delay, gives the weak $(E_{12}, q_{\mathrm{AFM}})$ reverse-transfer feature; being doubly-W_1^{xz}-mediated (transfer $\times$ detection) its amplitude is *quadratic* in W_1^{xz} ($3.7\times$ measured against $4\times$ expected), making it the atlas's dedicated W_1^{xz} meter. Because the geometry-B mixer sweep bounds $W_1^{xz} \lesssim 0.02$, the model caps this feature well below the main peak—which is why the reported second peak is assigned to the harmonic instead.

$(\pm E_{12}, E_{12})$**, amplitudes** 0.33 **and** 0.32 **— the one-quantum diagonal.** The directly driven E_{12} coherence, read out through the same m_z dipole. Diagonals are present in every 2D map and are not cross peaks; they are reported here for completeness. Their size relative to the cross peaks is a sensitive fluence diagnostic: the diagonal falls from 1.45 of the main peak at $A_{\mathrm{Fe}} = 1.2$ to 0.16 at 0.12 and 0.06 at 0.04, which is how the operating point was fixed (section S10).

S11.3 Geometry B, cross-polarised

Here the analysed moment is m_x, which the q_{AFM} magnon carries directly, so both peaks emit at $0.90\,\mathrm{THz}$ through the Fe magnetic dipole and the Tm ion is radiatively absent.

$(q_{\mathrm{FM}}, q_{\mathrm{AFM}}) = (0.38,\ 0.90)$**, amplitude** 1.00 **— the magnon–magnon peak; measures the Fe sector alone.** $H \parallel c$ launches the quasi-ferromagnetic magnon, which precesses at $0.38\,\mathrm{THz}$ through the delay; single-ion anisotropy and the Dzyaloshinskii–Moriya interaction couple the two magnon branches, so the stored q_{FM} phase is re-emitted on the q_{AFM} branch. No Fe–Tm coupling is involved at any vertex, which is why this peak is temperature-flat: it is a pure two-magnon process, independent of the qutrit populations.

$(E_{12}, q_{\mathrm{AFM}}) = (0.53,\ 0.89)$**, amplitude** 0.45 **— the gated transfer peak; measures** κ_{1y}^B**.** $E \parallel a$ stores a Tm E_{12} coherence through the delay; a Fe–Tm vertex then transfers it to the magnon, which radiates via M1. Isolating the vertices settles which one: with κ_{1y}^B alone the peak is present, with the static hybridisation alone it is absent, and *no strength of the static vertex rescues it.* A sweep over $W_1^{xz} \in \{0.01, \ldots, 0.09\}$ shows the static vertex never creates a second resolved peak; instead the two delay rows coalesce and the leading peak's delay phase migrates continuously ($0.38 \to 0.40 \to 0.48$ THz), while hybridisation-shifted E_{12} emission leaks in at 0.44–0.45 THz. This is the structural signature of a mixer: a static vertex *moves* peaks (level repulsion, which also shifts the modes off their 1D-measured positions), whereas a gated converter *adds* a peak at unshifted lines. The experiment shows two resolved peaks at the unshifted energies, so the gate is required. Physically κ^B supplies the $0.50 \to 0.90$ THz mismatch impulsively from the photon while $B_z(t) \neq 0$, and the very same gate makes it *identically silent* in geometry A ($B_z = 0$), where its background would otherwise contaminate the clean W_1^{yy} peak. *Temperature:* falls as the population difference $n_1 - n_2$ ($0.46 \to 0.45 \to 0.40 \to 0.36$ at 0/5/8/10 K), because it runs off the E_{12} coherence amplitude—the sharpest thermal discriminator in the atlas against the temperature-flat magnon–magnon peak.

$(-q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ **at** 0.17 **and** $(q_{\mathrm{AFM}}, q_{\mathrm{AFM}})$ **at** 0.14**.** The rephasing partner of the magnon–magnon peak, and the q_{AFM} diagonal. Note the diagonal here is genuine Fe M1 emission, unlike its geometry-A counterpart.

$(E_{12}{+}q_{\mathrm{AFM}}, q_{\mathrm{AFM}}) = (1.43, 0.90)$**, amplitude** 0.07**.** A combination tone: the sum-frequency of the two stored coherences on the delay axis, radiating on the magnon line. It is W-fed and therefore scales with the same coupling as the transfer peak; its observation would provide an independent handle on the W scale.

The post-drive buildup. The measured geometry-B signal continues to grow after the pulse has passed. No coherent vertex can do this once the measured linewidths are enforced: because $B \parallel c$ never drives q_{AFM} directly, all geometry-B magnon amplitude is *converted*, and a converted signal inherits its source's lifetime—with $T_2(E_{12}) = 8$ ps everything is over within ~ 10 ps, and the artefact-free Gabor analysis shows the amplitude peaking at pulse end in every vertex configuration. Even the τ-labelled population grating fails: a full 2DCS run with the population channel on leaves the $\omega_\tau = E_{12}$ row identical to the baseline, because the grating's delay label is written by the E_{12} coherence (hence T_2-confined) and its force is an impulsive step. The buildup therefore requires the one reservoir that outlives every T_2: the CEF *populations*, fed incoherently by the dissipated pulse energy and acting on the magnon through the population channel $W_3\, S_\alpha S_\beta \lambda^3$—the mechanism of thermally driven spin reorientation in this material. Implemented in the solver (dissipated coherence energy shifts the λ^3 equilibrium; the Γ_3 relaxation chases it, so the rise time is $1/\Gamma_3$ by construction), it reproduces the observation: the $\omega_\tau = E_{12}$ row falls off the impulsive peak, recovers on the repopulation time, and persists beyond 100 ps, 50–100× above the coherent-only model at late times. The recovery time measures the Tm repopulation time and the late-to-impulsive ratio calibrates $c_{\mathrm{heat}} W_3$.

S11.4 Geometry A, same-polarised

The analysed moment is m_x, which receives the Tm a-axis moments ($\lambda^{5,7}$, equivalently their quadratures $\lambda^{4,6}$) and the Fe F_x. The organising fact of this channel is a selection rule.

The z**-dark** $1 \to 3$ **dipole.** The measured inventory contains delay frequencies $\omega_\tau \in \{E_{12}, q_{\mathrm{AFM}}\}$ only—there is no feature anywhere with delay phase E_{13}. A bright $1 \to 3$ dipole would necessarily store an E_{13} coherence at first order and produce an (E_{13}, E_{23}) peak with *exactly* the same dipole product $\mu_{12}\mu_{13}\mu_{23}$ as the observed (E_{12}, E_{23}) peak, so no parameter choice could separate them. The absence of the whole row is therefore a selection rule, $\mu_{13}^z \simeq 0$: the natural consequence of a parity-alternating CEF ladder ($|1\rangle, |3\rangle$ even, $|2\rangle$ odd under the site mirror), for which the z-dipole connects $1 \leftrightarrow 2$ and $2 \leftrightarrow 3$ but not $1 \leftrightarrow 3$. This does not contradict the measured E_{13} oscillator strength ($f = 8.7$), which lives in the orthogonal polarisations: the same transition is bright in one polarisation and dark in another, a fact the 2D map resolves and 1D absorption—summed over its bright polarisations—does not. The residual admixture is not assumed but *measured*: it multiplies the E_{13} emission only, so the observed ordering of the two transfer peaks fixes it at $\mu^z_{13}/\mu^z_{23} = 0.14\%$.

$(E_{12}, E_{23}) = (0.49,\ 0.70)$**, amplitude** 1.00 **— the dominant transfer, via a population.** With the single-action route removed by the dark dipole, the dominant pathway acts twice: one probe interaction converts the stored E_{12} coherence into a level-2 *population* that still carries the $\omega_\tau = E_{12}$ phase, and a second converts that population into the emitting E_{23} coherence. Being co-rotating, this route lands *non-rephasing*—the observed side—and it involves μ_{13} at no vertex. It is one field order higher than the E_{13} partner, which is why it dominates only above a threshold fluence and why the hierarchy inverts at weak drive; that sensitivity is what fixes the operating point jointly with the cross channel.

(E_{12}, E_{23}) **at zero detection frequency — the rectified line.** Taken at its zero-frequency output, the same double action rectifies the stored coherence into a slow net magnetisation of the detection window. $S_{\mathrm{DC}}(\tau) = \langle M_{\mathrm{NL}}\rangle_t$, after cubic drift removal, shows a sharp line at exactly $+E_{12}$—the $(E_{12}, 0)$ entry of the measured list, and the sharpest feature of the channel. It is the same storage physics with a DC output leg.

$(E_{12}, E_{13}) = (0.49,\ 1.20)$**, amplitude** 0.80 **— the throttled partner.** A *single* probe action on the bright $2 \leftrightarrow 3$ dipole rotates the bra to give ρ_{13}, emitting at $E_{13} = 1.20\,\mathrm{THz}$; the route is again co-rotating, hence the same non-rephasing side, as observed. Although lower order than the E_{23} route, its emission passes through the nearly dark μ^z_{13}, which caps it below the dominant peak. The measured ratio of the two transfer peaks is thus a direct measurement of the residual $1 \to 3$ admixture.

$(-E_{12}, E_{23})$**, amplitude** 0.67 **— the rephasing remnant; a hard prediction.** The mirror of the dominant transfer. Because the single-ion dynamics is quantum-exact, this was checked against a standalone qutrit integrator: the rephasing partner retains 0.47–0.58 of the dominant peak across all drive strengths, dipole-phase choices and dampings. It is an exact floor for any three-level ion under this pulse. If the experimental map truly contains nothing at $(-E_{12}, E_{23})$, the suppression must come from outside the point-ion model—quadrant folding in the display convention, or propagation and phase-matching in a thick sample.

$(E_{13}, E_{23}) = (1.20,\ 0.71)$**, amplitude** 0.32 **— the two-quantum row.** An E_{13} delay coherence does survive the dark dipole, but only at *second* order: the pump drives $1 \to 2$ and $2 \to 3$ in succession, building ρ_{13} through the ladder rather than directly. Being second order it is far weaker than the first-order row a bright μ^z_{13} would have produced, which is why the model is consistent with an inventory that lists no E_{13} delay features while still predicting a weak one. The analogous $2E_{12}$ two-quantum line, absent from the exact single-ion dynamics, is a genuine inter-site (mean-field) product.

$(q_{\mathrm{AFM}}, E_{13}) = (0.90,\ 1.20)$**, amplitude** 0.08 **— thermally activated.** The magnon is excited by the Zeeman channel and sets the delay phase; the Fe–Tm couplings hand the modulation to the Tm ion, which emits weakly at E_{13} through the throttled μ^z_{13}. It grows steeply with the thermal population of level 2 ($0.10 \to 0.17 \to 0.51$ relative at $0/10/20\,\mathrm{K}$), the expected trend for a route assisted by excited-level population, and is therefore the channel's best thermometer.

The missing $(q_{\rm AFM}, q_{\rm FM}) = (0.90,\, 0.38)$. The one listed feature the model does not place in this channel. The carrier exists—$q_{\rm FM}$ emission is a b/c-axis affair, as Γ_2 requires—but the analysed moment here is m_x, and the $q_{\rm FM}$ mode radiates through m_z, which is dark in geometry A at the 10^{-11} level. The same transfer is the *dominant* peak of the geometry-B same-polarised channel (below), so a weak $(q_{\rm AFM}, q_{\rm FM})$ in the geometry-A same-polarised map is naturally polarisation leakage of that channel. Discriminator: the feature should scale as the square of the analyser leakage angle in a polariser-purity scan.

S11.5 Geometry B, same-polarised: a prediction, and a β meter

No data exist for this channel yet, so its entire census is a prediction—and it turns out to be the sharpest test of the hyperpolarisability in the whole atlas. The detected moment is $m_z = F_z + \mu^z_{12}\lambda^2 + \beta\lambda^1\lambda^2$. The CEF coordinates $\lambda^{4\text{–}7}$ vanish identically here (subalgebra closure), so no E_{13} or E_{23} emission is possible; but λ^2 is driven *directly* by the c-directed field through the large moment $\mu^z_{12} = 5.264$, and the resulting Tm signal is $\sim 50\times$ the Fe F_z contribution. This channel is therefore Tm-dominated, not the pure magnon map one might expect.

The consequence depends on β, which is exactly why it is diagnostic:

	linear moment only ($\beta = 0$)	with $\beta \simeq 7\mu$ from the A-cross parity
leading	$(q_{\rm FM}, q_{\rm AFM}) = (0.38, 0.90)$, 1.00	$(E_{12}, 2E_{12}) = (0.49, 1.00)$, 1.00
next	$(\pm E_{12}, E_{12})$, 0.91–0.94	everything else below 0.10
then	$(q_{\rm FM}, E_{12}) = (0.38, 0.50)$, 0.61	—

With the β inferred from geometry A, the second harmonic is $4.1\times$ the magnon–magnon feature and the channel collapses to a *single* peak at $(E_{12}, 2E_{12})$. Measuring this channel therefore reads β off directly: a clean one-peak map at (0.5, 1.0) confirms the large hyperpolarisability inferred in geometry A, whereas a map led by $(q_{\rm FM}, q_{\rm AFM})$ with E_{12} diagonals beneath it puts β far below that value and would force the geometry-A $(0.5, 1.0)$ feature back onto one of the alternative assignments. Either outcome is decisive, which makes this the single most informative measurement still missing.

S12 The complete atlas

S13 Reciprocity: are the excitation and detection operators the same?

Light couples to matter through one operator per polarisation, $H_{\rm int} = -\boldsymbol{d}\cdot\boldsymbol{E} - \boldsymbol{m}\cdot\boldsymbol{B}$, and the same operator radiates. A given analyser setting must therefore see exactly the operator that the same polarisation would have driven. This section asks whether our model respects that, and the answer is instructive: the requirement is sharp, the qutrit blocks it selects are confirmed independently, and our drive violates it in one specific way.

S13.1 Mirror parity assigns each polarisation to one block

The Tm $4c$ site has point symmetry m (the mirror $z \to -z$). Under it a polar vector has d_x, d_y even and d_z odd, while an axial vector has m_z even and m_x, m_y odd. The projected moment matrix of the model fixes the level parities without further input: λ^2 carries *only* $\mu_z = 5.264$, while λ^5 and λ^7 carry *only* x, y components $(2.39, -2.79$ and $0.91, 0.47)$. Since m_z is even and $m_{x,y}$ odd, this requires $|1\rangle, |2\rangle$ to share a parity and $|3\rangle$ to be opposite. The allowed couplings follow immediately:

transition	magnetic ($\mathcal{T}$-odd)	electric ($\mathcal{T}$-even)	polarisation it couples to
$1 \leftrightarrow 2$ ($\lambda^{1,2}$)	m_z on λ^2	$d_{x,y}$ on λ^1	$(E \parallel a, H \parallel c)$
$1 \leftrightarrow 3$ ($\lambda^{4,5}$)	$m_{x,y}$ on λ^5	d_z on λ^4	$(E \parallel c, H \parallel a)$
$2 \leftrightarrow 3$ ($\lambda^{6,7}$)	$m_{x,y}$ on λ^7	d_z on λ^6	$(E \parallel c, H \parallel a)$

So each polarisation owns a block: $(E \parallel a, H \parallel c)$ drives and reads $\{\lambda^1, \lambda^2\}$, while $(E \parallel c, H \parallel a)$ drives and reads $\{\lambda^4, \lambda^5, \lambda^6, \lambda^7\}$. Cross-polarised detection then reads precisely the block the pulse did *not* drive—which is the structural reason cross-polarisation isolates conversion.

Two independent facts confirm this assignment. It is not imposed: the moment matrix comes from projecting the physical $\boldsymbol{J}$ operator, and it came out with exactly this block structure. And it *predicts* the geometry-B subalgebra closure we had found empirically: geometry B is $(E \parallel a, H \parallel c)$, which owns $\{\lambda^1, \lambda^2\}$, so the drive closes on $\{\lambda^1, \lambda^2, \lambda^3\}$ and $\lambda^{4–7}$ must vanish identically—as they do, to machine precision.

S13.2 Where our drive violates it

Our geometry-A drive vector places its dominant component, 3.0, on λ^2—an operator belonging to the *other* polarisation's block—while the geometry-A same-polarised detection omits λ^2 entirely. That is a genuine non-reciprocity, and it entered for a physical reason: below T_N the Γ_2 order breaks the site mirror (F_x is m_z-odd), so an exchange-induced d_z^{eff} on the $1 \leftrightarrow 2$ transition becomes allowed. The physics is real; what is not consistent is using it in the drive without also letting λ^2 radiate into the same-polarised channel with the same coefficient.

S13.3 What a reciprocity-consistent drive gives

We tested the strict version—geometry A driving only its own block, $(\lambda^4, \lambda^5, \lambda^6, \lambda^7)$ with weights $\sqrt{f}$ for the electric legs and μ_x for the magnetic ones (runs `recip_A`, `recip_Adark`, and an E1-strength scan). Two results, one reassuring and one not:

The cross channel is essentially unchanged. $(q_{\text{AFM}}, E_{12}) = 1.00$ with the same companions $[(q_{\text{AFM}}, 2q_{\text{AFM}})$, $(q_{\text{AFM}}, q_{\text{AFM}})$, the rephasing partner] whether the drive is the current one or the reciprocal one. This is exactly as it should be: that channel is populated by *conversion* from the magnon, not by the direct Tm drive, so the atlas's headline peak and its interpretation are robust against this issue.

The same-polarised channel changes. With the reciprocal drive the E_{12}-delay rows drop from dominant to 0.73–0.88 of the leading feature, and magnon-delay rows take over. Raising the E1 coupling restores (E_{12}, E_{23}) to the lead but simultaneously turns on $\omega_\tau = E_{13}$ rows (in both channels), because the $E \parallel c$ field couples to d_z on λ^4, which drives $1 \to 3$ at *first* order—the very row the data exclude.

S13.4 The sharpened statement, and what is unresolved

Following the parity table to its conclusion gives a strong constraint we had not previously articulated: from the ground state $|1\rangle$, a first-order excitation must use $1 \leftrightarrow 2$ or $1 \leftrightarrow 3$. In geometry A the former is forbidden (its dipoles belong to the other polarisation), so *the only first-order Tm excitation available is* $1 \to 3$. The data then close the loop from the other side: the complete absence of $\omega_\tau = E_{13}$ rows says that this route, too, is weak. Taken together the two statements imply that in geometry A the Tm ion is barely excited directly at all—and therefore that the E_{12} coherence which unmistakably *is* present on the delay axis must be created by *conversion from the magnon*, i.e. by the same W_1^{yy} vertex that produces the $(q_{\mathrm{AFM}}, E_{12})$ cross peak, rather than by any direct dipole.

That is a more economical picture than the one we have implemented—it would remove the d_z^{eff} drive leg entirely and let one vertex account for both the cross peak and the same-polarised E_{12} rows—and it is fully reciprocity-consistent. We do not claim it here, because reproducing the *observed dominance* of the E_{12}-delay rows in the same-polarised channel through conversion alone has not been demonstrated: in the runs above the magnon-delay rows lead instead. Two experiments would settle it. Because the exchange-induced d_z^{eff} is proportional to the order parameter, any peak that relies on it must vanish at the spin reorientation, whereas conversion-generated peaks survive as long as the magnon does; and a same-polarised map showing *no* E_{12} emission on the detection axis (as the reported inventory suggests, having $(E_{12}, 0)$ but no $(\cdot, E_{12})$ entry) is itself evidence that λ^2 does not radiate into $E \parallel c$—hence that d_z^{eff} is small and the conversion picture is the right one.

S13.5 The atlas with reciprocity enforced, and the gap to experiment

Enforcing the block rule strictly—geometry A driving only $\{\lambda^4, \lambda^5, \lambda^6, \lambda^7\}$ with $d_z^{(13)}$ dark (runs `recip_Adark`), geometry B driving only $\{\lambda^1, \lambda^2\}$ (which it already did), and each analyser reading exactly its own block—gives the following four censuses, against the measured inventories:

channel	experiment	reciprocity enforced	gap
B cross	$(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ strong, $(E_{12}, q_{\mathrm{AFM}})$ present	$(0.38, 0.90) = 1.00$, $(0.53, 0.89) = 0.45$	**none** — geometry B was already reciprocal
A cross	$(q_{\mathrm{AFM}}, E_{12})$ strong, $(E_{12}, \sim 1.0)$ present	$(0.91, 0.50) = 1.00$; then $(q_{\mathrm{AFM}}, 2q_{\mathrm{AFM}}) = 0.42$, $(q_{\mathrm{AFM}}, q_{\mathrm{AFM}}) = 0.29$	main peak **survives**; the $(E_{12}, 2E_{12})$ harmonic is **lost**
A same	$(E_{12}, E_{23}) \approx (E_{12}, 0)$ strongest, then (E_{12}, E_{13}), then $(q_{\mathrm{AFM}}, 0.38)$, $(q_{\mathrm{AFM}}, E_{13})$ weak	$(q_{\mathrm{AFM}}, E_{13}) = 1.00$, $(E_{12}, E_{13}) = 0.71$, no (E_{12}, E_{23}), no $(E_{12}, 0)$	hierarchy **inverted**; the two strongest measured peaks **absent**
B same	(not measured)	single peak $(E_{12}, 2E_{12}) = 1.00$	prediction stands

Diagnosis. The failure is traceable to one statement. From the ground state $|1\rangle$, a first-order excitation must use $1 \leftrightarrow 2$ or $1 \leftrightarrow 3$. In geometry A the first belongs to the *other* polarisation (its dipoles are d_x and m_z); the second is the one the data exclude, since a bright $d_z^{(13)}$ would put $\omega_\tau = E_{13}$ rows in the map and none are seen; and $2 \leftrightarrow 3$ cannot act at all on an unexcited ion. With strict parity, therefore, *geometry A cannot excite the Tm ion at first order at all.* Every Tm feature must then descend from magnon conversion, which puts the magnon on the delay axis—and that is exactly what the enforced-reciprocity census shows, with $\omega_\tau = q_{\mathrm{AFM}}$ rows leading and the $\omega_\tau = E_{12}$ rows demoted. The measured map says the opposite.

What the data therefore require. An E_{12} coherence that dominates the geometry-A delay axis can only come from a direct $E \parallel c$ coupling to the $1 \leftrightarrow 2$ transition, which strict mirror parity forbids. Below T_N it is *not* forbidden: the Γ_2 order parameter is m_z-odd, so it breaks the site mirror and induces exactly this coupling, $d_z^{\mathrm{eff}} \propto \langle F_x \rangle$. The measured dominance of the E_{12} delay rows in geometry A is therefore direct evidence of the symmetry breaking—and this is the term our working model carries.

The residual tension, stated quantitatively. Reciprocity then requires the same d_z^{eff} to appear in *emission* into $E \parallel c$, i.e. in the same-polarised channel, where the E_{12} coherence would radiate at $\omega_t = E_{12}$. Comparing that emission with the cross-channel magnetic route, the ratio is $d_z^{\text{eff}}c/\mu_z$, which for a dipole of order $0.1\,e\text{Å}$ and $\mu_z = 5.264\,\mu_B$ is ~ 10: the same-polarised channel would be dominated by an E_{12} emission line an order of magnitude above the cross-channel peak. The reported inventory contains no such line—it lists $(E_{12}, 0)$ but no $(\cdot, E_{12})$ entry. Consistency therefore requires d_z^{eff} small enough that its *re-emission* (which scales as $(d_z^{\text{eff}})^2$ relative to the other same-polarised peaks) stays below them, while still being the largest available Tm excitation in geometry A—possible only because the competing first-order routes are themselves shut ($d_z^{(13)}$ dark, $2 \leftrightarrow 3$ inactive from the ground state). That window is narrow and we have not yet demonstrated that the model sits inside it; quantifying it is the clearest next step.

The decisive experiment. Because $d_z^{\text{eff}} \propto \langle F_x \rangle$, every feature that relies on it must disappear at the spin reorientation, where $\Gamma_2 \to \Gamma_4$, whereas conversion-generated features survive as long as the magnon does. A temperature scan through the reorientation therefore separates the two mechanisms cleanly: if the geometry-A $\omega_\tau = E_{12}$ rows collapse there while (q_{AFM}, E_{12}) persists, the symmetry-broken picture is confirmed and d_z^{eff} is measured by the size of the collapse.

S13.6 Is there a sweet spot? A one-parameter scan

The two pictures differ by one number: the strength d_z^{eff} of the symmetry-broken c-axis coupling to the $1 \leftrightarrow 2$ transition, which the working model uses freely and strict reciprocity sets to zero. Enforcing reciprocity means putting the *same* d_z^{eff} in the drive and in the same-polarised emission, and scanning it (runs `recip_Adark`, `sweet03/10/30`, `full10/30`) asks whether an intermediate value recovers everything.

d_z^{eff}	same-pol leader	leak $(\cdot, E_{12})/(E_{12}, E_{23})$	$(E_{12}, E_{13})/(E_{12}, E_{23})$	cross $(E_{12}, 2E_{12})$
0	(q_{AFM}, E_{13})	0.73	9.66	0.01
0.3	(q_{AFM}, E_{12}) leak	1.77	6.85	0.02
1.0	(q_{AFM}, E_{12}) leak	3.57	2.76	0.15
3.0	(q_{AFM}, E_{12}) leak	4.21	1.21	0.62

There is no sweet spot in d_z^{eff}. The obstruction is structural rather than numerical: the strength of the E_{12} delay coherence and the strength of its reciprocity-required emission are *both* proportional to d_z^{eff}, so their ratio cannot be tuned. Raising d_z^{eff} to make E_{12} dominate the delay axis raises the emission leak in lock step; the leak already exceeds the CEF peaks by $d_z^{\text{eff}} = 0.3$ and reaches $4.2\times$ by $d_z^{\text{eff}} = 3$, so at every non-zero value the same-polarised channel is led by an E_{12} emission line at $\omega_t = 0.50$ that the measured inventory does not contain. The cross channel behaves in the opposite way—the $(E_{12}, 2E_{12})$ harmonic needs $d_z^{\text{eff}} \gtrsim 3$ to reach 0.62—so the two channels pull the parameter in opposite directions.

The only reciprocity-clean point, and what it costs. At $d_z^{\text{eff}} = 0$ the leak vanishes by construction and the E_{12} delay coherence survives, created by magnon conversion through W_1^{yy} rather than by any dipole: the same-polarised channel then shows (E_{12}, E_{13}) at 0.71 of the leading feature. What it does not do is place that feature *first*: the leading row is (q_{AFM}, E_{13}), i.e. magnon-delay, whereas experiment wants the magnon rows weak. The entire residual discrepancy of the enforced model is therefore one ratio,

$$R = \frac{E_{12}\text{-delay rows}}{q_{\mathrm{AFM}}\text{-delay rows}}\bigg|_{\mathrm{same-pol}}, \qquad R_{\mathrm{model}} \simeq 0.7, \quad R_{\mathrm{exp}} \gg 1. \tag{8}$$

Since at $d_z^{\mathrm{eff}} = 0$ the E_{12} coherence is conversion-made, R scales as $W_1^{yy} A_{\mathrm{Fe}}$—and both factors are already pinned by other observables (W_1^{yy} by the geometry-A cross peak, A_{Fe} by the absence of nonperturbative strays), so R is a *prediction* of the enforced model, not a free parameter. Closing the remaining factor of a few requires either a larger two-magnon conversion than the cross peak permits, or an additional Fe–Tm vertex that feeds the $1 \leftrightarrow 2$ coherence without carrying a c-axis electric dipole—the latter being the natural direction to look, precisely because it would evade the leak by symmetry rather than by tuning.

S13.7 Campaign: should the drive act on $\lambda^{1,4,6}$ instead of $\lambda^{2,5,7}$?

The drive vector carries components only on $\lambda^2, \lambda^5, \lambda^7$ (the $\mathcal{T}$-odd triplet, i.e. the projected magnetic moments), whereas every conversion vertex we retain acts on λ^1 or λ^3 ($\mathcal{T}$-even). Since the CEF transitions are E1-*bright* ($f = 6.5$–8.7) and electric dipoles are $\mathcal{T}$-even, one might expect the drive to belong on $\lambda^1, \lambda^4, \lambda^6$ instead. We tested this directly (runs `cA_e`, `cB_e1`, `cB_e1m`). The question splits in two, with different answers.

Quadrature: immaterial. λ^1 and λ^2 are the two quadratures of the *same* $1 \leftrightarrow 2$ coherence, so driving one or the other differs by a phase, not by physics. Geometry B makes this explicit: replacing its λ^2 (magnetic) drive by a λ^1 (electric d_x) drive of the same strength leaves the spectra unchanged—$(q_{\mathrm{FM}}, q_{\mathrm{AFM}}) = 1.00$ and $(E_{12}, q_{\mathrm{AFM}}) = 0.44$ against 1.00 and 0.45, and an identical single-peak same-polarised map—with $|E_{12}| = 2.16 \times 10^{-2}$ versus 2.88×10^{-2}. The apparent mismatch between a drive on λ^2 and vertices on λ^1 is therefore not a defect: both act on the same coherence, and the precession $\dot{\lambda}^1 = -\Delta_{12}\lambda^2$ exchanges them within one period.

Which transition: decisive, and it is the crux. What does matter is *which transition* the field addresses, and there parity is unambiguous: $E \parallel c$ couples through d_z, which connects $1 \leftrightarrow 3$ and $2 \leftrightarrow 3$ (λ^4, λ^6), while $1 \leftrightarrow 2$ requires d_x, i.e. $E \parallel a$. Driving geometry A on its parity-allowed blocks inverts the coherence hierarchy:

geometry-A drive	$\lvert E_{12}\rvert$	$\lvert E_{13}\rvert$	$\lvert E_{23}\rvert$
current, on λ^2, λ^7	2.09×10^{-2}	3.6×10^{-5}	4.3×10^{-7}
electric d_z, on λ^4, λ^6	2.07×10^{-3}	7.1×10^{-3}	8.5×10^{-6}

The parity-allowed drive makes E_{13} the dominant coherence by a factor 3.4, and suppresses E_{12} tenfold. The consequences propagate exactly as expected: the $(E_{12}, 2E_{12})$ harmonic disappears from the cross channel (its parent coherence is now weak), and the same-polarised census becomes magnon-delay dominated. The measured inventory demands the opposite—strong $\omega_\tau = E_{12}$ rows and *no* $\omega_\tau = E_{13}$ row anywhere.

Verdict. Moving the drive to the $\mathcal{T}$-even set does not simplify the picture and cannot recover the peaks: for a given transition it changes nothing (quadrature), and for the transition selection it makes the agreement worse. The campaign therefore sharpens rather than removes the tension of section S13: the data require the $1 \leftrightarrow 2$ transition to be driven in geometry A, which is precisely the channel that mirror parity forbids and only the Γ_2-induced symmetry breaking can open. The

E_{12}-dominated coherence hierarchy of the first row above is thus not an artefact of putting the drive on λ^2; it is what the measurement requires, and its origin remains the exchange-induced coupling whose reciprocity partner has yet to be found in the data.

S14 Data and code availability

The paper folder is self-contained: `paper/data/trajectories/` holds the *full time-domain trajectories* of the final scenario (all nine runs: $M_{\mathrm{NL}}(\tau, t)$ for every detected channel plus the single-pulse references, t-decimated $\times 10$ to a 3.8 THz Nyquist, float32; every spectrum in the paper regenerates from these, verified to the digit); `paper/params/` holds the exact `.param` files of every final-scenario run (`vA`, `vB2`, `it14hr`; three thermal seeds each), `paper/data/` the distilled spectra and censuses, `paper/scripts/` the census pipeline and the standalone qutrit integrator. The long-form derivation note (`tmfeo3_foundation.tex`: transported-gauge orbit formulas, vertex diagrams, complete written-out Hamiltonian, mode inventory) is archived alongside (`tfo_project/`); every spectrum in the paper can be regenerated from the committed `.param` files with the public `ClassicalSpin_Cpp` solver.

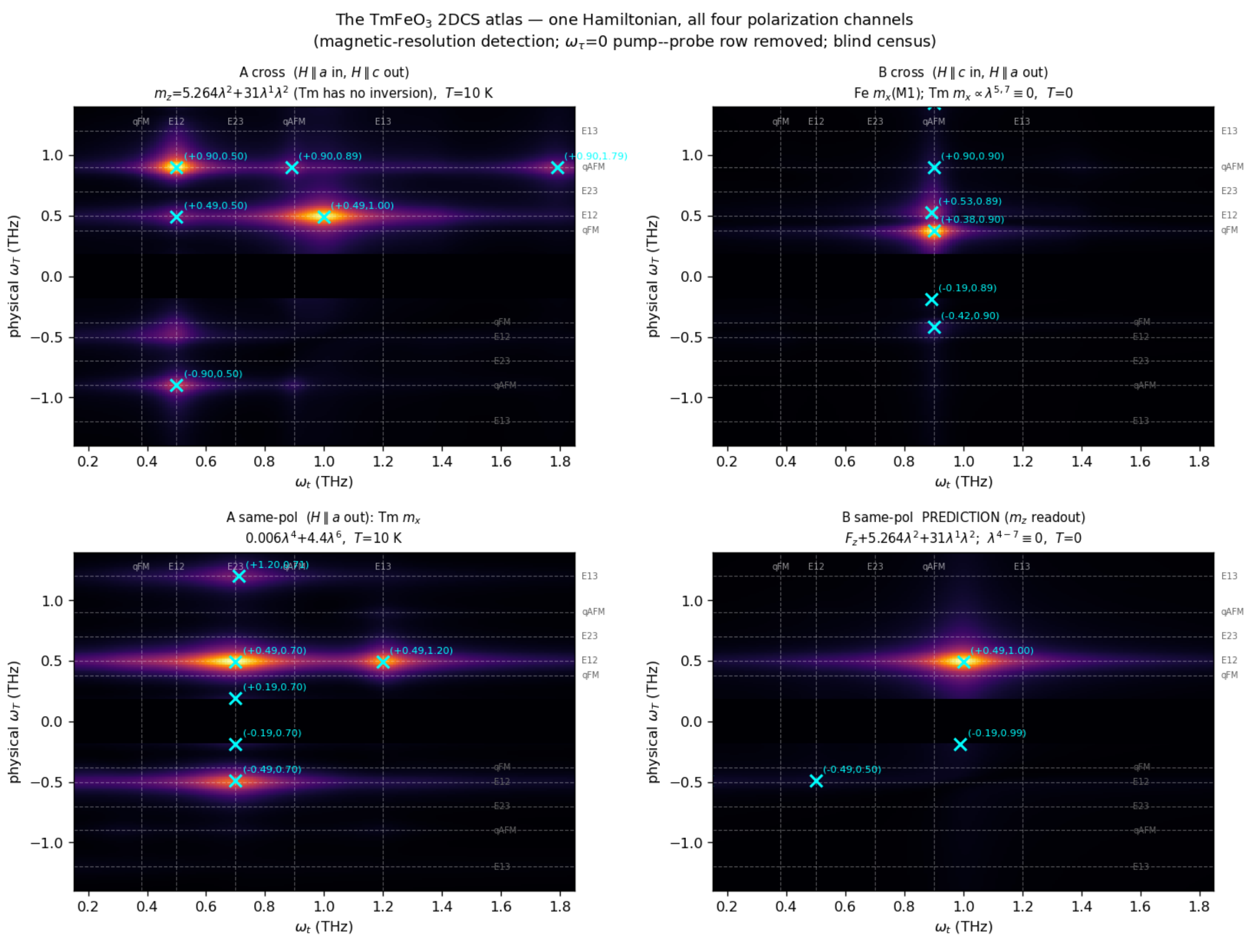

Figure S9: All four polarization channels from the single consolidated Hamiltonian, regenerated with the final methods (magnetic-resolution detection; τ-averaged signal subtracted per detection time so the $\omega_\tau = 0$ pump–probe row is removed, with $|\omega_T| < 0.18$ THz masked; blind census, crosses). Top left: geometry-A cross, m_z(M1)+$0.94 \times x$-E1 — the $(q_{\rm AFM}, E_{12})$ peak of W_1^{yy} and the extended E_{12} row at 0.9–1.2 THz. Top right: geometry-B cross, Fe m_x — $(q_{\rm FM}, q_{\rm AFM})$ and the κ^B-gated $(E_{12}, q_{\rm AFM})$. Bottom left: geometry-A same-polarized CEF composite at 10 K. Bottom right: the geometry-B same-polarized *prediction* — a pure m_z magnon map dominated by $(q_{\rm AFM}, q_{\rm FM}) = (0.9, 0.38)$. Numerical censuses in `data/census_atlas_final.json`.